Fixed Income Analysis – Exam preperations

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List of Corrections

1 Fixed Income Securities

A fixed income security is an investment that provides a return in the form of fixed periodic interest payments and the eventual return of principal at maturity. The payments of a fixed income security are known in advance. Bonds are the most common form of fixed income securities.

A derivative is an instrument which value is derived from a primary instrument (stocks, currency, bonds, other derivatives etc.) also called underlying instrument. Derivative contracts divided into OTC-derivatives (over the counter), stock exchange derivatives. Derivatives are priced using Martingale pricing theory. Derivatives are effective tools for controlling risk. With them you can separate risk and trade it separately. Enabeling traders to sell unwanted risk or buy risk to diversify the portfolio.

1.1 Forward contract

A forward contract is an agreement between a buyer and a seller to trade an underlying asset at a future date for a price specified at the contract's inception. In order to have a single transaction, the price of a forward is such that the value is zero for both parties (Most common: Currency forwards, forward rate agreements.)

Pricing

A forward with maturity T and delivery price K provides a payoff of $P_T - K$ at time T and no other payments. Here, P is the underlying (typically the price of an asset or a specific interest rate). In an arbitrage free market, there exists an equivalent martingale measure $\mathbb{Q}$ (associated with the money market numeraire), such that the time $t < T$ value of this payoff is:

$$V_t = \mathbb{E}_t^{\mathbb{Q}}\left[e^{-\int_t^T r_u \, du}(P_T - K)\right] = \mathbb{E}_t^{\mathbb{Q}}\left[e^{-\int_t^T r_u \, du} P_T\right] - K B_t^T,$$

where B_t^T denotes the time t price of a zero-coupon bond which pays 1 at maturity T.

For $t \leq T$, the forward price is set to the value of K that solves $V_t = 0$, since this implies that neither party must make an upfront payment. Thus,

$$F_t^T = \frac{\mathbb{E}_t^{\mathbb{Q}}\left[e^{-\int_t^T r_u \, du} P_T\right]}{B_t^T}$$

Under the risk-neutral measure $\mathbb{Q}$, the discounted asset price is a martingale. So for an asset with no intermediate income before T we have $F_t^T = \frac{P_t}{B_t^T}$.

1.2 European options

European options provide the right, but not the obligation to trade the underlying asset on expiration date for a pre-specified (at time t) price K (American options can be exercised on any date). A Call is the right to buy, and a Put is the right to sell the asset.

Pricing

Consider a call option with expiration date T and strike K on the underlying P_t. The payoff at time T (and hence the call price at T) is

$$C_T = \max\{P_T - K, 0\}$$

Again, in an arbitrage free market, there exists an equivalent martingale measure $\mathbb{Q}$, such that the discounted asset prices are martingales.

Assuming that the underlying and zero-coupon bonds are traded, we may change the $\mathbb{Q}$-measure to the T-forward martingale measure $\mathbb{Q}^T$, which uses B_t^T as numeraire. Using this measure we can derive a general pricing formula for a European call option:

$$\begin{aligned}
C_t &= B_t^T \mathbb{E}_t^{\mathbb{Q}^T}\left[\max\{P_T - K, 0\}\right] \\
&= B_t^T \mathbb{E}_t^{\mathbb{Q}^T}\left[\mathbb{1}_{\{P_T>K\}}(P_T - K)\right] \\
&= B_t^T \left(\mathbb{E}_t^{\mathbb{Q}^T}\left[\mathbb{1}_{\{P_T>K\}}P_T\right] - K\mathbb{Q}_t^T(P_T > K)\right) \\
&= B_t^T \mathbb{E}_t^{\mathbb{Q}^T}\left[\mathbb{1}_{\{P_T>K\}}P_T\right] - KB_t^T\mathbb{Q}_t^T(P_T > K) \\
\text{(Change of numeraire)} \quad &= \mathbb{E}_t^{\mathbb{Q}^P}\left[B_t^T P_T \frac{B_T^T}{P_T}\frac{P_t}{B_t^T}\mathbb{1}_{\{P_T>K\}}\right] - KB_t^T\mathbb{Q}_t^T(P_T > K) \\
\text{(Measurable)} \quad &= P_t\mathbb{E}_t^{\mathbb{Q}^P}\left[\mathbb{1}_{\{P_T>K\}}\right] - KB_t^T\mathbb{Q}_t^T(P_T > K) \\
&= P_t\mathbb{Q}_t^P(P_T > K) - KB_t^T\mathbb{Q}_t^T(P_T > K)
\end{aligned}$$

where $\mathbb{Q}^P$ is the martingale measure using P_t as numeraire, thus we have assumed that P_t is also traded in the market. The proof is completely analogous for European puts, only we use the payoff $\max\{K - P_T, 0\}$. Note that the final probabilities correspond to the option being in the money under the two different equivalent martingale measures. To compute these probabilities further assumptions of the market model is needed, for instance if we set

$$\mathbb{Q}_t^P(P_T > K) = N(d_1) \quad \text{and} \quad \mathbb{Q}_t^T(P_T > K) = N(d_2),$$

where $N(\cdot)$ denotes the standard normal cumulative distribution function. The general pricing formula reduces to the Black-Scholes call option price.

1.3 Relevant results

Bonds, Yields and Rates

FiXme Note: Fill out this section

Other Fixed-Income securities

Interest rate forwards, forward rate agreements (FRAs)

FiXme Note: Add pricing formulas

Future

Like a Forward, a Futures contract is a contract where the two parties agree to trade the underlying at a perspeciefed strike. The main difference is that futures are marked-to-market, at least daily. This means that the two counterparties continuously settle the difference between the previous futures price and the new future price.

Thus, you are implicitly exposed to the current interest rate, through the future price. Whereas, with a forward you only take into account the expected future rate. Thus, the difference in price between the two contracts is the discounted covariance between the zero-coupon bond and the payment price at time T. A Futures contract has value:

$$\phi_t^T = \mathbb{E}^{\mathbb{Q}}[P_T].$$

Futures on bonds

Options on bonds

Caps

A cap protects a floating-rate borrower against the risk of paying very high interest rates. Work as a ceiling on the interest rate, compensating the holder if the rate rises above a specified level. Consists of a series of underlying options called caplets (call option on the spot interest rate).

Floors

A floor protects a floating-rate lender against the risk of receiving very low interest rates. Works like caps, consists of floorlets (put option on floating interest rate with delayed payment of the payoff).

Collars

A combination of caps and floors. Ensures that the interest rate payments on a floating rate borrowing arrangement stay between two prespecified levels. (Can be seen as a portfolio of a long position in a cap with rate K_c and a short position in floor with rate K_f.)

Swaps

An interest rate swap is an exchange of two cash flow streams determined by certain interest rates. A plain vanilla interest rate swap takes two forms:

- Payer swap: Pays a stream of fixed rate payments and recieves a stream of floating rate payments.
- Receiver swap: Pays a stream of floating rate payments and receives a stream of fixed rate payments.

Forward swap: Agreement to enter into a swap with a future starting date and fixed rate already set.

Forward swaps

An agreement to enter a swap at an agreed-upon future starting date T_0 and fixed rate. The forward swap rate $\tilde{L}_t^{\delta,T_0}$ is the fixed rate that solves $P_t = 0$ at time $t \leq T$.

Swaptions

A European swaption gives its holder the right at the expiry date to enter into an interest rate swap that has a given fixed rate. A payer swaption gives the right to enter into a payer swap, and likewise for receiver swaption.

Repurchase agreements

Short term contracts used for lending money based on high-grade securities as collateral (government bonds). Low interest rate, but also (a bit) risky, as is might be difficult to renew loans in a distressed economy (think fall 2008, illuquidity).

- Parties agree on Prices P_0 and $P_T = P_0(1 + rT)$
- At time $t = 0$, the security is traded at price P_0, and the security repurchased at time T, using price P_T

The collateral reduces the lender's exposure to counter party (credit) ris. Repos are subject to daily margin calls if collateral value changes (like futures)

Financial crisis: some bank could not "repay" the repo due to a cut in funding by another bank (Goldman Sachs, GS), so not enough capital to do so. GS bought credit default swaps a few days before, so probably knew what was going to happen.

FiXme Note: Maybe a section on the different martingale measures we have seen

2 One-factor diffusion models

All one-factor models assume that the short rate is the only state variable. That is, r_t contains all relevant information about the term structure. Assume the short rate follows a diffusion process

$$\mathrm{d}r_t = \alpha(r_t)\ \mathrm{d}t + \beta(r_t)\ \mathrm{d}z_t$$

where $(z_t)_{t\geq 0}$ is a one-dimensional standard Brownian motion (under the real-world measure $\mathbb{P}$). The short rate dynamics under the risk-neutral measure $\mathbb{Q}$ is

$$\mathrm{d}r_t = \hat{\alpha}(r_t)\ \mathrm{d}t + \beta(r_t)\ \mathrm{d}z_t^{\mathbb{Q}}$$

where $\hat{\alpha}(r_t) = \alpha(r_t) - \beta(r_t)\lambda(r_t)$, and where λ is the market price of risk (risk premium per unit volatility).

Note that these models are called *time homogeneous* since $\hat{\alpha}, \beta$ do not depend explicitly on calendar time t.

2.1 Affine models

A model is affine if the risk-neutral drift and the variance coefficients are affine functions of the state variable, r_t, i.e.

$$\hat{\alpha}(r_t) = \hat{\varphi} - \hat{\kappa} r_t, \quad \text{and} \quad \beta(r)^2 = \delta_1 + \delta_2 r_t \geq 0, \tag{2.1}$$

where $\hat{\varphi}, \hat{\kappa}, \delta_1, \delta_2$ are constants. When $\hat{\alpha}$ and β^2 are linear in r we can split the PDE into two ODEs.

ZCB Prices in Affine Models

Theorem 7.1 (Zero-Coupon Bond Prices)
In an affine model, zero-coupon bond prices are given by

$$B_t^T = B^T(r_t, t) = \mathrm{e}^{-a(T-t) - b(T-t) r_t},$$

where the functions $a(\tau)$ and $b(\tau)$ solve $a(0) = b(0) = 0$ and

$$\frac{1}{2}\delta_2 b(\tau)^2 + \hat{\kappa} b(\tau) + b'(\tau) - 1 = 0 \tag{2.2}$$

$$a'(\tau) - \hat{\varphi} b(\tau) + \frac{1}{2}\delta_1 b(\tau)^2 = 0, \quad \tau > 0$$

These ODEs are known as Riccati equations. For certain parameters, a and b can be found explicitly.

Proof
We prove that the candidate for a solution, is a solution of the fundamental PDE and then use that under standard regulariy the boundary value problem has a unique solution. The fundamental PDE of $B^T(r,t)$ is ($\mathcal{S}$ is state space of r):

$$\frac{\partial B^T(r,t)}{\partial t} + \hat{\alpha}(r)\frac{\partial B^T(r,t)}{\partial r} + \frac{1}{2}\beta(r)^2 \frac{\partial^2 B^T(r,t)}{\partial r^2} - rB^T(r,t) = 0, \tag{2.3}$$
$$B^T(r,T) = 1, \quad (r,t) \in \mathcal{S} \times [0,T].$$

The terminal condition is obviously satisfied since

$$B^T(r,T) = \mathrm{e}^{-a(0) - b(0)r} = 1$$

Next, we find the derivatives

$$\frac{\partial B^T(r,t)}{\partial t} = B^T(r,t)(a'(T-t) + b'(T-t)r),$$
$$\frac{\partial B^T(r,t)}{\partial r} = -B^T(r,t)b(T-t),$$
$$\frac{\partial^2 B^T(r,t)}{\partial r^2} = B^T(r,t)b(T-t)^2.$$

Inserting in Equation (2.3) and dividing by $B^T(r,t)$ we obtain

$$a'(T-t)+b'(T-t)r-\hat{\alpha}(r)b(T-t)+\frac{1}{2}\beta(r)^2b(T-t)^2-r=0.$$

Using Equation (2.1) and gathering terms we find

$$\begin{aligned}&a'(T-t)+b'(T-t)r-(\hat{\varphi}-\hat{\kappa}r)b(T-t)+\frac{1}{2}(\delta_1+\delta_2 r)\,b(T-t)^2-r=0\\ \Rightarrow\quad &a'(T-t)-\hat{\varphi}b(T-t)+\frac{1}{2}\delta_1 b(T-t)^2+\left(b'(T-t)+\hat{\kappa}b(T-t)+\frac{1}{2}\delta_2 b(T-t)^2-1\right)r=0,\end{aligned}$$

for all $(r,t)\in\mathcal{S}\times[0,T]$. Inserting $\tau=T-t$ we observe that the above holds when Equation (2.2) holds.

Q.E.D.

2.2 Varsicek's model

In Varsicek's model we assume that the short rate r_t is driven by an Ornstein-Uhlenbeck process under $\mathbb{P}$:

$$\mathrm{d}r_t=\kappa\left(\theta-r_t\right)\,\mathrm{d}t+\beta\,\mathrm{d}z_t$$

where κ,θ and β are positive constants. Given the market price of risk $\lambda(r,t)$, the $\mathbb{Q}$-dynamics are

$$\mathrm{d}r_t=\left[\kappa\left(\theta-r_t\right)-\beta\lambda(r,t)\right]\,\mathrm{d}t+\beta\,\mathrm{d}z_t^{\mathbb{Q}}=\kappa\left(\hat{\theta}-r_t\right)\,\mathrm{d}t+\beta\,\mathrm{d}z_t^{\mathbb{Q}}$$

Assuming $\lambda(r,t)=\lambda$, i.e. constant market price of risk, we have $\hat{\theta}=\theta-\frac{\beta\lambda}{\kappa}$.

Properties

r is also an Ornstein–Uhlenbeck process under $\mathbb{Q}$, but in this case it is mean-reverting around $\hat{\theta}$. The model is Gaussian, implying that r_T can take negative values with positive probability, which is often regarded as a drawback of the model.

The model furthermore assumes constant volatility. It is an affine term structure model with parameters

$$\hat{\varphi}=\kappa\hat{\theta},\quad \hat{\kappa}=\kappa,\quad \delta_1=\beta^2,\qquad \delta_2=0,$$

and therefore the price of a zero-coupon bond is given by

$$B^T(r,t)=\mathrm{e}^{-a(T-t)-b(T-t)r},$$

where the equations in Equation (2.2) can be solved to:

$$b(\tau)=\frac{1}{\kappa}\left(1-\mathrm{e}^{-\kappa\tau}\right)\qquad\text{and}\qquad a(\tau)=\left(\hat{\theta}-\frac{\beta^2}{2\kappa^2}\right)(\tau-b(\tau))+\frac{\beta^2}{4\kappa}b(\tau)^2$$

Relevant results

Zero-Coupon yields

For affine and Varsicek the yield curve satisfies

$$y^T(r,t)=-\frac{\ln\left(B^T(r,t)\right)}{T-t}=\frac{a(T-t)}{T-t}+\frac{b(T-t)}{T-t}r.$$

Affine function of the short rate. For $b(\tau)>0$ yields are increasing in the short rate, an increase in r shifts the curve upwards (unless $b(\tau)$ proportional to τ), prices decreasing in the short rate. Not all shapes are possible, cannot generate trough and twists.

The instantaneous forward rates (at time T) for affine model (maybe Varsicek) is

$$f^T(r,t)=-\frac{\frac{\partial B^T(r,t)}{\partial T}}{B^T(r,t)}=a'(T-t)+b'(T-t)r$$

Cox-Ingersoll-Ross (CIR) model

Market price of risk is $\frac{\lambda\sqrt{r}}{\beta}$, $\mathbb{Q}$-dynamics is square root proces

$$\mathrm{d}r_t = \hat{\kappa}\left(\hat{\theta} - r_t\right)\,\mathrm{d}t + \beta\sqrt{r_t}\,\mathrm{d}z_t^{\mathbb{Q}}$$

Mean reverting (around $\hat{\theta}$). Non-negative interest rates, 0 is reflecting barrier. Affine with $\hat{\varphi} = \hat{\kappa}\hat{\theta}, \hat{\kappa} = \hat{\kappa}, \delta_1 = 0$ and $\delta_2 = \beta^2$ (ODEs can be solved). Time varying volatility (increases with level).

Generalized affine models

In Varsicek we assume the market price of risk is constant. Can let it be affine function of r to allow for more flexibility. Still affine. Same way of thinking can be used for CIR.

Non Affine models

- Quadratic models: Dynamics $r_t = x_t^2$ (x_t Ornstein-Uhlenbeck process). a and b as in affine case. 0 is refelcting barrier (disadvantage). $B^T(x,t) = \mathrm{e}^{-a(T-t)-b(T-t)x-c(T-t)x^2}$. Complicated closed form expressions for prices of derivatives.
- Shadow-rate models: Dynamics $r_t = \max\{0, x_t\}, x_t$ OU process. Only nonnegative values. 0 is absorbing barrier (occurs in practice). Bond and derivative prices computed numerically.

3 Multi-factor diffusion models

One-factor models are not perfect as they cannot generate all the shapes of empirically observed yield curves. Specifically they cannot generate, they cannot generate trough and twists of the yield curve and any changes in two yields or bond prices will be perfectly correlated (change in short yield affects long yield).

In this presentation I will introduce the concept of Multi-factor models try to accommodate this problem, by adding more state variables, which can affect the model both independent of, or dependent on the short rate. We will consider the subclass of Affine Multi-factor models and describe how one can price zero-coupon bonds using the model.

3.1 Affine Multi-factor models

We first construct the multi-factor models. Let $\boldsymbol{x}_t = (x_{1t}, x_{2t}, \ldots, x_{nt})^T$ be a vector of state variables, and assume that these have affine $\mathbb{Q}$-dynamics given by:

$$\mathrm{d}\boldsymbol{x}_t = \Big(\underset{n\times 1}{\hat{\varphi}} - \underset{n\times n}{\hat{\kappa}} \underset{n\times 1}{\boldsymbol{x}_t} \Big)\, \mathrm{d}t + \underset{n\times n}{\Gamma} \sqrt{\underset{n\times n}{\boldsymbol{V}(\boldsymbol{x}_t)}}\, \underset{n\times 1}{\mathrm{d}\boldsymbol{z}_t^{\mathbb{Q}}},$$

where $\boldsymbol{V}(\boldsymbol{x}_t)$ is the diagonal matrix

$$\boldsymbol{V}(\boldsymbol{x}_t) = \begin{bmatrix} v_{0,1} + \mathbf{v}_{1,1}^T \boldsymbol{x}_t & 0 & \cdots & 0 \\ 0 & v_{0,2} + \mathbf{v}_{1,2}^T \boldsymbol{x}_t & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & v_{0,n} + \mathbf{v}_{1,n}^T \boldsymbol{x}_t \end{bmatrix}$$

For admissibility we require $v_{0,i} + \mathbf{v}_{1,i}^T \boldsymbol{x}_t \geq 0$ for all i to ensure that the square root is well-defined, and $\boldsymbol{x}_t \in \mathbb{R}^n$. We further assume that the short rate is affine in the state variables, $\boldsymbol{x}_t$:

$$r_t = \xi_0 + \xi^T \boldsymbol{x}_t.$$

And we assume that the market price of risk is given by

$$\lambda(\boldsymbol{x}_t) = \sqrt{\boldsymbol{V}(\boldsymbol{x}_t)}\bar{\lambda},$$

for some vector $\bar{\lambda} \in \mathbb{R}^n$. Such that the $\mathbb{P}$-dynamics take the (same) affine form:

$$\mathrm{d}\boldsymbol{x}_t = (\varphi - \kappa \boldsymbol{x}_t)\ \mathrm{d}t + \Gamma \sqrt{\boldsymbol{V}(\boldsymbol{x}_t)}\, \mathrm{d}\boldsymbol{z}_t$$

where $\varphi = \hat{\varphi} + \Gamma\psi, \kappa = \hat{\kappa} - \Gamma\Psi$ with

$$\psi = \begin{bmatrix} \bar{\lambda}_1 v_{0,1} \\ \bar{\lambda}_2 v_{0,2} \\ \vdots \\ \bar{\lambda}_n v_{0,n} \end{bmatrix}, \quad \text{and} \quad \Psi = \begin{bmatrix} \bar{\lambda}_1 \mathbf{v}_{1,1}^T \\ \bar{\lambda}_2 \mathbf{v}_{1,2}^T \\ \vdots \\ \bar{\lambda}_n \mathbf{v}_{1,n}^T \end{bmatrix}$$

3.2 Pricing and characterization

In the model setup above, the price of a zero-coupon bond is given by

$$B_t^T = B^T(x_t, t) = \mathrm{e}^{-a(T-t) - b(T-t)^T x_t}$$

where the functions $a(\tau)$ and $b(\tau)$ solve the Riccati ODE's, i.e. $a(0) = b(0) = 0$ and

$$b'(\tau) = -\hat{\kappa}^T b(\tau) - \frac{1}{2} \sum_{i=1}^{n} \left[\Gamma^T b(\tau)\right]_i^2 v_{1,i} + \xi$$

$$a'(\tau) = \hat{\varphi}^T b(\tau) - \frac{1}{2} \sum_{i=1}^{n} \left[\Gamma^T b(\tau)\right]_i^2 v_{0,i} + \xi_0$$

> **Note**
> There are $n+1$ ODE's, to solve these first find $b(\tau)$ from the first equality and then integrate the second to obtain $a(\tau)$.

Affine Multi-factor models of size $n \in \mathbb{N}$ can be divided into $n+1$ categories, denoted by $\mathbb{A}_m(n)$ for each $m \in \{0,1,\ldots,n\}$. For a fixed m each class is characterized by:

- The first m factors affect the volatility $\boldsymbol{V}(\boldsymbol{x}_t)$, and are known as volatility factors.
- The remaining $n-m$ do not affect $\boldsymbol{V}(\boldsymbol{x}_t)$ and are known as non-volatility factors. Vasicek is $\mathbb{A}_0(1)$ model, CIR is $\mathbb{A}_1(1)$. We need to be careful about the admissibility requirement.

Parameter restrictions for volatility factors

For a volatility factor x_i where $i \in 1,\ldots,m$, to ensure admissibility we need x_i to be non-negative, hence we impose the following conditions:

- $\varphi_i > 0, \quad \kappa_{ii} > 0,$ for mean reversion,
- $\kappa_{ij} = 0$, for $j = m+1,\ldots,n, \quad \kappa_{ij} \leq 0$, for $j = 1,\ldots,m$, with $j \neq i$,
- Furthermore, x_i must only be sensitive to $\mathrm{d}z_{it}$ with volatility coefficient proportional to $\sqrt{x_{it}}$.

This leads to

$$\mathrm{d}x_{it} = \left(\varphi_i - \sum_{j=1}^{n} \kappa_{ij} x_{jt}\right) \mathrm{d}t + \Gamma_{ii}\sqrt{x_{it}}\, \mathrm{d}z_{it}.$$

Thus, volatility factors are instantaneously uncorrelated square root processes. But they are not necessarily independent because volatility factors affect each others drifts (positively). Also, the volatility factors cannot be dependent on the non-volatlity factors (they may be negative).

Parameter restrictions for non-volatility factors

For the non-volatility factors x_i, to ensure admissibility:

- Drift rate may depend on all factors.
- The sensitivity to the volatility shocks $\mathrm{d}z_{jt}$ must be proportional to $\sqrt{x_{jt}}$ $(j = 1,2,\ldots,m)$.
- The sensitivity to non-volatility shocks $\mathrm{d}z_{jt}$ $(j = m+1,\ldots,n)$ is

$$\Gamma_{ii}\sqrt{v_{0,j} + \sum_{k=1}^{m} v_{1,jk} x_{kt}},$$

  with $v_{0,j}, v_{1,jk} \geq 0$.

This leads to

$$\mathrm{d}x_{it} = \left(\varphi_i - \sum_{j=1}^{n} \kappa_{ij} x_{jt}\right) \mathrm{d}t + \sum_{j=1}^{m} \Gamma_{ij}\sqrt{x_{jt}}\, \mathrm{d}z_{jt} + \sum_{j=m+1}^{n} \Gamma_{ij}\sqrt{v_{0,j} + \sum_{k=1}^{m} v_{1,jk} x_{kt}}\, \mathrm{d}z_{jt}.$$

Relevant results

Principal component analysis with US bond data shows that first factor (level) explains about 85% of the variation in yields, second factor (slope) about 10%, third factor (curvature) about 2%. So we need at least two factors. So generally, there is marginal return in model accuracy compared to complexity if we add more than three factors.

Generalized affine models

The idea is to allow a general market price of risk, which gives more flexible setting when estimating model parameters

Completely affine models

$$\begin{aligned} r_t &= \xi_0 + \xi^T x_t \\ \mathrm{d}x_t &= (\hat{\varphi} - \hat{\kappa} x_t)\ \mathrm{d}t + \Gamma \sqrt{\boldsymbol{V}(x_t)}\ \mathrm{d}z_t^{\mathbb{Q}} \\ \lambda(x_t) &= \sqrt{\boldsymbol{V}(x_t)}\bar{\lambda} \end{aligned}$$

The market price of risk is affected only by volatility factors.

Essentially affine models

$$\lambda_t = \sqrt{\boldsymbol{V}(x_t)}\bar{\lambda}_1 + \sqrt{\boldsymbol{V}(x_t)^{-}}\bar{\lambda}_2 x_t$$

where $\boldsymbol{V}(x_t)^{-}$is diagonal matrix (broadly speaking the inverse of $\boldsymbol{V}(x_t)$). Here, the non-valotility factors can also affect the market price of risk.

Extended affine models

$$\sqrt{\boldsymbol{V}(x_t)}\lambda_t = \bar{\lambda}_1 + \bar{\lambda}_2 x_t$$

with restrictions in $\bar{\lambda}_2$ to ensure that the process for x_t is well-defined under $\mathbb{Q}$ and $\mathbb{P}$. Offers more flexibility for the volatility factors on the market price of risk.

4 Estimation

The formulas in this course rely on some model parameters, which in practice need to be estimated. For this, we consider Kalman filter based qausi maximum likelihood (QML). The Kalman filter is used to update the distribution of the state variables x_t and QML is used to maximize the log-likelihood. We have two cases: Observed x_t and unobserved x_t.

Observed

Assume x_t has Vasicek $\mathbb{P}$-dynamics

$$dx_t = \kappa\left(\theta - x_t\right)\ dt + \beta\ dz_t$$

and we have sample $\{x_1, \ldots, x_T\}$. Simple MLE can be applied (since it's Gaussian). Typically we do not observe x_t, if we for instance try to model the short rate. We can also only estimate $\mathbb{P}$-model parameters and not the market price of risk.

Unobserved

Assume x_t has Vasicek $\mathbb{P}$-dynamics

$$dx_t = \kappa\left(\theta - x_t\right)\ dt + \beta\ dz_t$$

In the multi-factor affine model we have the yield

$$y_t(\tau) = A_\tau + B_\tau' x_t$$

and we observe $y_t(\tau)$ for $t \in \{1, \ldots, T\}$ and $\tau \in \{\tau_1, \ldots, \tau_N\}$. A Kalman filter is then used to update the distribution of x_t. In particular from $t = 1$ to $t = T$ we sequentially update mean and variance of Gaussian x_t as we observe new y_t. Then find parameters by maximizing log-likelihood (QML). This way, we can estimate both $\mathbb{P}$ and $\mathbb{Q}$ parameters and market price of risk λ.

4.1 State space system

Linear state space system has two equations, measurement and transition

$$\begin{aligned} y_t &= A + Bx_t + \epsilon_t, \quad \epsilon_t \sim N(0, H) \\ x_t &= C + Dx_{t-1} + \eta_t, \quad \eta_t \sim N(0, Q) \end{aligned}$$

The measurement equation is similar to the affine equations between yields and state variables. If we have N maturities of yields and k state variables A is a $N \times 1$ matrix and B is $N \times k . \epsilon_t$ is measurement error (deviations of observed data from model implied prices). We use $\mathbb{Q}$-dynamics of x_t.

The transition equation describes the dynamics of the state variables only. C is $k \times 1, D$ and Q is $k \times k$. η_t is innovation error. Here, we use the $\mathbb{P}$-dynamics of x_t.

4.2 Kalman filter estimation

Will consider Kalman filter estimation for the unobserved case. We assume Gaussian distribution of x_t and update it sequentially. It contains three steps: Initialization, prediction and update.

Initialization step

The initial distribution is

$$x_0 \sim N\left(x_{0|0}, \Sigma_{0|0}\right)$$

where $x_{0|0} = \mathbb{E}[x_0]$ and $\Sigma_{0|0} = \mathbb{V}[x_0]$ are unconditional mean and variance of stationary state variable x_t.

Prediction step

In this step we predict x_t given observations available at time $t-1$, i.e. $Y_{t-1} = \{y_1, \ldots, y_{t-1}\}$. From the previous point in time we have

$$x_{t-1} \mid Y_{t-1} \sim N\left(x_{t-1|t-1}, \Sigma_{t-1|t-1}\right)$$

The transition equation implies

$$x_t \mid Y_{t-1} \sim N\left(x_{t|t-1}, \Sigma_{t|t-1}\right)$$

where

$$\begin{aligned} x_{t|t-1} &= \mathbb{E}[x_t \mid Y_{t-1}] = C + D\mathbb{E}[x_{t-1} \mid Y_{t-1}] = C + Dx_{t-1|t-1} \\ \Sigma_{t|t-1} &= \mathbb{V}[x_t \mid Y_{t-1}] = D\mathbb{V}[x_{t-1} \mid Y_{t-1}]D' + Q = D\Sigma_{t-1|t-1}D' + Q \end{aligned}$$

Update step

Here, we update x_t given the additional observation y_t, so information set is $Y_t = \{y_1, \ldots, y_t\}$. From the prediction step we know that

$$x_t \mid Y_{t-1} \sim N\left(x_{t|t-1}, \Sigma_{t|t-1}\right)$$

We update the distribution of x_t as

$$x_t \mid Y_t \sim N\left(x_{t|t}, \Sigma_{t|t}\right)$$

where

$$\begin{aligned} x_{t|t} &= \mathbb{E}[x_t \mid Y_t] \\ \Sigma_{t|t} &= \mathbb{V}\left[x_{t|t-1} \mid Y_t\right] \\ &= \Sigma_{t|t-1}B'F_t^{-1}v_t \\ &= \Sigma_{t|t-1}B'F_t^{-1}B\Sigma_{t|t-1} \end{aligned}$$

with

$$v_t = y_t - \mathbb{E}[y_t \mid Y_{t-1}], \quad \text{and} \quad F_t = \mathbb{V}[v_t \mid Y_{t-1}] = B\Sigma_{t|t-1}B' + H$$

(If y_t away from the mean the updated mean should go up. The data y_t is seen in v_t)

Relevant results

Maximum likelihood

Given discrete observations $\{y_1, \ldots, y_T\}$ (y_t a $N \times 1$ vector) we want to estimate parameters ψ. The joint density can be decomposed to

$$f(y_1, \ldots, y_T) = \prod_{t=2}^{T} f(y_t \mid y_1, \ldots, y_{t-1}) f(y_1)$$

Denoting $Y_{t-1} = \{y_1, \ldots, y_{t-1}\}$ and $Y_0 = \emptyset$, we choose ψ that maximizes

$$L(\psi) = \ln(f(y_1, \ldots, y_T; \psi)) = \sum_{t=1}^{T} \ln(f(y_t \mid Y_{t-1}))$$

For a Gaussian random vector this leads to

$$\ln(f(x)) = -\frac{N}{2}\ln(2\pi) - \frac{1}{2}\ln(|\Sigma|) - \frac{1}{2}(x-\mu)'\Sigma^{-1}(x-\mu)$$

Justification of results

The results are justified be the following result about multivariate normal distributions. Let

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim N(\mu, \Sigma)$$

where

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \text{and} \quad \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}.$$

Then

$$X_1 \mid X_2 = x_2 \sim N\left(\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{12}'\right).$$

5 Christensen, Diebold, Rudebusch (Affine arbitrage-free Nelson-Siegel models)

The usual Nelson-Siegel (NS) parameterization is simple to estimate, popular due to empirical success, but not dynamic and does not exclude arbitrage. Affine models theoretically appealing (arbitrage-free), not succesful empirically, estimation is nontrivial (lot of parameters).

CDR, Journal of Econometrics: "The affine arbitrage-free class of Nelson-Siegel term structure models." Combines the two approaches. Introduce new class of affine models that are arbitrage-free and has the yield-curve of the NS parameterization. Denoted AFNS (arbitrage-free Nelson-Siegel). Their empirical analysis involves in-sample fit and out-of-sample forecasting of flexible and restricted versions of AFNS models. Three main conclusions:

- Flexible versions do better in terms of in-sample fit.
- Restricted versions perform better for out-of-sample forecasting.
- Restricted versions beats the $\mathbb{A}_0(3)$ (0 volatility factors, 3 non-volatility factors) model of Duffee in forecasting.

Nelson-Siegel

The original NS parameterization of the yield curve is

$$y(\tau) = \beta_0 + \beta_1\left(\frac{1-\mathrm{e}^{-\lambda\tau}}{\lambda\tau}\right) + \beta_2\left(\frac{1-\mathrm{e}^{\lambda\tau}}{\lambda\tau} - \mathrm{e}^{-\lambda\tau}\right).$$

Interpretations of the four parameters:
$-\beta_0$ is a constant, hence affects long term yields. (level)
$-\beta_1$ affects short term rates since $\frac{1-\mathrm{e}^{-\lambda\tau}}{\lambda\tau}$ is a decreasing function that starts a 1 and decays to 0 . (slope)
$-\beta_2$ affects medium term rates since $\frac{1-\mathrm{e}^{\lambda\tau}}{\lambda\tau} - \mathrm{e}^{-\lambda\tau}$ starts at 0 , then increases and later decays to 0 . (curvature)
$-\lambda$ is a scaling factor that affects the exponential decay rate.

5.1 AFNS model

The $\mathbb{Q}$-dynamics of the state variable X_t is 3 -factor Gaussian

$$\mathrm{d}X_t = \kappa^{\mathbb{Q}}\left(\theta^{\mathbb{Q}} - X_t\right)\,\mathrm{d}t + \Sigma\,\mathrm{d}W_t^{\mathbb{Q}}.$$

We assume that $r_t = \rho_0 + \rho_1' X_t$. The yield curve is given by

$$y_t(\tau) = \frac{a(\tau)}{\tau} + \sum_{j=1}^{3}\frac{b_j(\tau)}{\tau}X_t^j,$$

where the simplified ODE of $b = (b_1, b_2, b_3)$ is

$$b'(\tau) = -\left(\kappa^{\mathbb{Q}}\right)' b(\tau) + \rho_1, \quad b(0) = 0$$

(the last term disappears since we have constant covariance matrix)
Theorem (main result for AFNS models). Let $\lambda > 0, r_t = X_t^1 + X_t^2$, where the dynamics of $\left(X_t^1, X_t^2, X_t^3\right)'$ are

$$\begin{pmatrix} \mathrm{d}X_t^1 \\ \mathrm{d}X_t^2 \\ \mathrm{d}X_t^3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \lambda & -\lambda \\ 0 & 0 & \lambda \end{pmatrix}\left[\begin{pmatrix} \theta_1^{\mathbb{Q}} \\ \theta_2^{\mathbb{Q}} \\ \theta_3^{\mathbb{Q}} \end{pmatrix} - \begin{pmatrix} X_t^1 \\ X_t^2 \\ X_t^3 \end{pmatrix}\right]\mathrm{d}t + \Sigma\begin{pmatrix} \mathrm{d}W_t^{1,\mathbb{Q}} \\ \mathrm{d}W_t^{2,\mathbb{Q}} \\ \mathrm{d}W_t^{3,\mathbb{Q}} \end{pmatrix}.$$

Then the yield curve is given by

$$y_t(\tau) = X_t^1 + \frac{1-\mathrm{e}^{-\lambda\tau}}{\lambda\tau}X_t^2 + \left(\frac{1-\mathrm{e}^{-\lambda\tau}}{\lambda\tau} - \mathrm{e}^{-\lambda\tau}\right)X_t^3 - \frac{A(\tau)}{\tau}$$

where A is the solution of a certain ODE.
Proof. The ODE $b'(\tau) = -\left(\kappa^{\mathbb{Q}}\right)' b(\tau) + \rho_1$ (here $\rho_1 = (1,1,0)'$) is equivalent to

$$\begin{aligned} b_1'(\tau) &= 1 \\ b_2'(\tau) &= -\lambda b_2(\tau) + 1 \\ b_3'(\tau) &= \lambda b_2(\tau) - \lambda b_3(\tau) \end{aligned}$$

Since $b(0) = 0$ we have

$$b_1(\tau) = \tau, \quad \text{and} \quad b_2(\tau) = \frac{1 - \mathrm{e}^{-\lambda\tau}}{\lambda}$$

Using the value of b_2 we obtain

$$b_3(\tau) = \frac{1 - \mathrm{e}^{-\lambda\tau}}{\lambda} - \tau \mathrm{e}^{-\lambda\tau}$$

Will skip the last factor (yield-adjustment term), but it is computed by plugging b_1, b_2, b_3 into the multifactor Ricatti ODE for A and integrating.

The above only considers the $\mathbb{Q}$-dynamics of X_t. For the $\mathbb{P}$-dynamics we use the essentially affine models from Duffee. In the Gaussian case we have $\lambda_t = \lambda_1 + \lambda_2 X_t$, so the $\mathbb{P}$-dynamics become

$$\mathrm{d}X_t = \kappa^{\mathbb{P}}\left(\theta^{\mathbb{P}} - X_t\right) \mathrm{d}t + \Sigma\, \mathrm{d}W_t^{\mathbb{P}}$$

Although the $\mathbb{Q}$-dynamics was restricted (e.g. zero terms in $\kappa^{\mathbb{Q}}$) we can choose (λ_1, λ_2) appropriately to get any $\kappa^{\mathbb{P}}$ and $\theta^{\mathbb{P}}$.

Additional stuff

5.2 Dynamic NS:

$$y_t(\tau) = L_t + S_t\left(\frac{1 - \mathrm{e}^{-\lambda\tau}}{\lambda\tau}\right) + C_t\left(\frac{1 - \mathrm{e}^{-\lambda\tau}}{\lambda\tau} - \mathrm{e}^{-\lambda\tau}\right)$$

where L_t, S_t, C_t represent level, slope and curvature factors ($= \beta_{0t}, \beta_{1t}, \beta_{2t}$). There are two DNS models: L_t, S_t, C_t first order autroregression and L_t, S_t, C_t first order vector autoregression. The DNS model has the transition equation

$$\begin{pmatrix} L_t - \mu_L \\ S_t - \mu_S \\ C_t - \mu_C \end{pmatrix} = A \begin{pmatrix} L_{t-1} - \mu_L \\ S_{t-1} - \mu_S \\ C_{t-1} - \mu_C \end{pmatrix} + \begin{pmatrix} \eta_t(L) \\ \eta_t(S) \\ \eta_t(C) \end{pmatrix}$$

where η are noises.
Estimations from article: The article estimates four models:

- Independent-factor AFNS (diagonal κ and diagonal Σ).
- Correlated-factor AFNS (full κ and lower triangular Σ).
- Independent-factor DNS (diagonal A and independent noises).
- Correlated-factor DNS (full A and correlated noises).

6 Kim and Singleton (Quadratic and shadow rate models)

The affine model class has been the workhorse for studying interest rates so far. Its tractability and richness in theory are some of the benefits. The recent financial crisis brought us into uncharted territory - near zero (or negative) rates for an extended period. So which models are suitable for dealing with near zero rates?

KS, Journal of Econometrics: "Term structure models and the zero-lower bound: An empirical investigation of Japanese yields." They study three classes of 2-factor models which imply non-negative interest rates: Affine multi-factor CIR model, quadratic model, shadow-rate model. Their analysis show that the latter two are very succesful in capturing the distribution of Japanese yields. Affine 2 -factor CIR performs worse.

6.1 Theory (for all models)

Under no arbitrage, the price of a τ-year zero-coupon bond is

$$P_\tau(x_t) = \mathbb{E}_t^{\mathbb{Q}}\left[\mathrm{e}^{-\int_t^{t+\tau} r(x_s)\,\mathrm{d}s}\right]$$

The dynamics of the state variable are

$$\mathrm{d}x_t = \mu^{\mathbb{Q}}(x_t)\ \mathrm{d}t + \Sigma(x_t)\ \mathrm{d}B_t^{\mathbb{Q}} = \left(\mu^{\mathbb{Q}}(x_t) + \Sigma(x_t)\lambda(x_t)\right)\ \mathrm{d}t + \Sigma(x_t)\ \mathrm{d}B_t^{\mathbb{P}}$$

The instantaneous yield volatility is

$$\sigma_{t,\tau} = \left(\frac{\partial y_\tau(x_t)}{\partial x_t'}\Sigma(x_t)\Sigma(x_t)'\frac{\partial y_\tau(x_t)}{\partial x_t}\right)^{\frac{1}{2}}.$$

The instantaneous expected excess return is

$$\rho_{t,\tau} = -\tau\frac{\partial y_\tau(x_t)}{\partial x_t'}\Sigma(x_t)\lambda(x_t)$$

Quadratic-Gaussian models

The 2 -factor quadratic-Gaussian model is given by

$$\begin{aligned} r_t &= x_t'\Phi x_t \\ \mathrm{d}x_t &= \kappa^{\mathbb{Q}}\left(\theta^{\mathbb{Q}} - x_t\right)\ \mathrm{d}t + \Sigma\ \mathrm{d}B_t^{\mathbb{Q}} \\ \lambda_t &= \lambda_a + \Lambda_b x_t \end{aligned}$$

where $\Phi, \kappa^{\mathbb{Q}}, \Sigma, \Lambda_b$ are 2×2 and $\theta^{\mathbb{Q}}, \lambda_a$ are 2×1 matrices. (Note: x_t is a Gaussian model, i.e. in $\mathbb{A}_0(2)$). The yield of maturity τ is

$$y_{t,\tau} = a_\tau + b_\tau' x_t + x_t' C_\tau x_t$$

To ensure non-negative r_t we assume that Φ is symmetric positive semi-definite, which means (note $D_{11}, D_{22} \geq 0$)

$$\Phi = ADA' = \begin{bmatrix} 1 & 0 \\ A_{21} & 1 \end{bmatrix}\begin{bmatrix} D_{11} & 0 \\ 0 & D_{22} \end{bmatrix}\begin{bmatrix} 1 & A_{21} \\ 0 & 1 \end{bmatrix}$$

This gives

$$r_t = D_{11}(x_{1t} + A_{21}x_{2t})^2 + D_{22}x_{2t}^2$$

If $D_{11}, D_{22} > 0$ we have

$$r_t = 0 \Longrightarrow x_{1t} = x_{2t} = 0 \Longrightarrow y_{t,\tau} = a_\tau,$$

which is constant and we want to avoid. So one of x_{1t}, x_{2t} should be free (non-zero). A solution is choosing $D_{22} = 0 \Longrightarrow x_{2t}$ is free. Short rate, yields volatilities and excess return satisfy

$$\begin{aligned}
\sigma_t &= 2\left(x_t'\Phi\Sigma\Sigma'\Phi x_t\right)^{\frac{1}{2}} \\
\sigma_{t,\tau} &= \left((b_\tau + 2C_\tau x_t)'\Sigma\Sigma'(b_\tau + 2C_\tau x_t)\right)^{\frac{1}{2}} \\
\rho_{t,\tau} &= -\tau\left(b_\tau + 2C_\tau x_t\right)'\Sigma\left(\lambda_a + \Lambda_b x_t\right)
\end{aligned}$$

Additionally, $r_t = 0$ implies $\sigma_t = 0$ and $\sigma_{t,\tau}, \rho_{t,\tau} \approx 0$ for small τ.

Shadow-rate models

Here,

$$r_t = \max\{s_t, 0\}$$

where s_t is some (shadow) stochastic process. When r_t touches zero it stays there for some time, hence zero is an absorbing barrier for r_t. For s_t we consider the following 3-factor model

$$\begin{aligned}
s_t &= \rho + \gamma' x_t + x_t'\Phi x_t \\
\mathrm{d}x_t &= \kappa^{\mathbb{Q}}\left(\theta^{\mathbb{Q}} - x_t\right)\,\mathrm{d}t + \Sigma\,\mathrm{d}B_t^{\mathbb{Q}} \\
\lambda_t &= \lambda_a + \Lambda_b x_t
\end{aligned}$$

where ρ is a constant and $\Phi, \kappa^{\mathbb{Q}}, \Sigma, \Lambda_b$ are 2×2 and $\gamma, \theta^{\mathbb{Q}}, \lambda_a$ are 2×1 matrices.
Letting $P(t,x)$ be the price of a zero-coupon bond maturing at T Munk inplies

$$\frac{\partial P}{\partial t} + \left(\kappa^{\mathbb{Q}}\left(\theta^{\mathbb{Q}} - x_t\right)\right)'\frac{\partial P}{\partial x} + \frac{1}{2}\operatorname{tr}\left(\frac{\partial^2 P}{\partial x^2}\Sigma\Sigma'\right) - \max\{s(x), 0\} = 0$$

with $P(T,x) = 1$ for all x. (By a change of variable, we get the PDE from the article). Due to the nonlinear term $\max\{s(x), 0\}$ we cannot solve the PDE explicitly (numerical methods include finite difference (KS use this), Monte Carlo, trees). No explicit formulas for volatilities and excess returns either. For understanding let's consider the 1 -factor model

$$\begin{aligned}
r_t &= \max\{s_t, 0\} \\
\mathrm{d}s_t &= \kappa^{\mathbb{P}}\left(\theta^{\mathbb{P}} - s_t\right)\,\mathrm{d}t + \sigma\,\mathrm{d}B_t^{\mathbb{P}} \\
\lambda_t &= \bar{\lambda}
\end{aligned}$$

If $s_t > 0$ then the short rate $\sigma_t = \sigma$ and if $s_t \leq 0$ then $\sigma_t = 0$, so there is a discontinuity in volatility at $s_t = 0$. We also have

$$\sigma_{t,\tau} = \frac{\partial y_\tau(s_t)}{\partial s_t}\sigma, \quad \rho_{t,\tau} = -\tau\frac{\partial y_\tau(s_t)}{\partial s_t}\sigma\bar{\lambda}$$

Additional comments: When $s_t = 0$ we have $r_t = 0$ and $\frac{\partial y_\tau(s_t)}{\partial s_t} > 0$. Thus, $\sigma_{t,\tau}$ and $\rho_{t,\tau}$ stay away from 0 even for small τ. Also, when $s_t \to -\infty$ we have $r_t = 0$ and $\frac{\partial y_\tau(s_t)}{\partial s_t}$ tends to 0 . When $s_t \to \infty$ the term $\frac{\partial y_\tau(s_t)}{\partial s_t}$ becomes constant.

6.2 Additional stuff

Affine models: The 2 -factor CIR model ($\mathbb{A}_2(2)$) with $\mathbb{Q}$-dynamics

$$\begin{bmatrix} \mathrm{d}x_{1t} \\ \mathrm{d}x_{2t} \end{bmatrix} = \begin{bmatrix} \kappa_{11}^{\mathbb{Q}} & \kappa_{12}^{\mathbb{Q}} \\ \kappa_{21}^{\mathbb{Q}} & \kappa_{22}^{\mathbb{Q}} \end{bmatrix}\left(\begin{bmatrix} \theta_1^{\mathbb{Q}} \\ \theta_2^{\mathbb{Q}} \end{bmatrix} - \begin{bmatrix} x_{1t} \\ x_{2t} \end{bmatrix}\right)\mathrm{d}t + \begin{bmatrix} \sqrt{x_{1t}} & 0 \\ 0 & \sqrt{x_{2t}} \end{bmatrix}\begin{bmatrix} \mathrm{d}B_{1t}^{\mathbb{Q}} \\ \mathrm{d}B_{2t}^{\mathbb{Q}} \end{bmatrix},$$

where

$$\left[\kappa^{\mathbb{Q}}\theta^{\mathbb{Q}}\right]_i > 0, \quad \kappa_{ii}^{\mathbb{Q}} > 0, \quad i = 1,2, \quad \text{and} \quad \kappa_{12}^{\mathbb{Q}}, \kappa_{21}^{\mathbb{Q}} < 0$$

Assume

$$r_t = \gamma_1 x_{1t} + \gamma_2 x_{2t}, \quad \text{with} \quad \gamma_1, \gamma_2 \geq 0$$

The yield of a τ-year bond is

$$y_{t,\tau} = a_\tau + b_{1,\tau} x_{1t} + b_{2,\tau} x_{2t}$$

where

$$a_\tau = \frac{a(\tau)}{\tau}, \quad b_{1,\tau} = b_{1,\tau} = \frac{b_1(\tau)}{\tau}, \quad b_{2,\tau} = \frac{b_2(\tau)}{\tau}.$$

Consider to specifications for the market price of risk (completely affine and extended affine)

$$\lambda_{ca} = \begin{bmatrix} \Lambda_{11}\sqrt{x_{1t}} \\ \Lambda_{22}\sqrt{x_{2t}} \end{bmatrix}, \quad \lambda_{ea} = \begin{bmatrix} (\lambda_1 + \Lambda_{11} x_{1t} + \Lambda_{12} x_{2t})/\sqrt{x_{1t}} \\ (\lambda_2 + \Lambda_{21} x_{1t} + \Lambda_{22} x_{2t})/\sqrt{x_{2t}} \end{bmatrix}$$

Short rate and yield volatilities are

$$\sigma_t = \left(\gamma_1^2 x_{1t} + \gamma_2^2 x_{2t}\right)^{\frac{1}{2}}$$
$$\sigma_{t,\tau} = \left(b_{1,\tau}^2 x_{1t} + b_{2,\tau} x_{2t}\right)^{\frac{1}{2}}$$

Expected excess returns are given by

$$\rho_{t,\tau,ca} = -\tau\left(b_{1,\tau}\Lambda_{11} x_{1t} + b_{2,\tau}\Lambda_{22} x_{2t}\right)$$
$$\rho_{t,\tau,ea} = -\tau\left[b_{1,\tau}\left(\lambda_1 + \Lambda_{11} x_{1t} + \Lambda_{12} x_{2t}\right) + b_{2,\tau}\left(\lambda_2 + \Lambda_{21} x_{1t} + \Lambda_{22} x_{2t}\right)\right]$$

7 Calibration of diffusion models

With time-homogeneous models the yield curve y_t^τ only depends on maturity (which is reasonable). However, we cannot perfectly match the observed yield curve (undesirable fact for derivatives trading). Therefore, we will focus on the time-inhomogeneous models

$$\mathrm{d}r_t = \hat{\alpha}\left(r_t, t\right)\ \mathrm{d}t + \hat{\beta}\left(r_t, t\right)\ \mathrm{d}z_t^{\mathbb{Q}}$$

specifically the affine subset where

$$\hat{\alpha}\left(r_t, t\right) = \hat{\varphi}(t) - \hat{\kappa}(t) r_t, \quad \hat{\beta}\left(r_t, t\right) = \sqrt{\delta_1(t) + \delta_2(t) r_t}$$

The pricing function for a zero-coupon bond is given by

$$B^T\left(r_t, t\right) = \mathrm{e}^{-a(t,T) - b(t,T) r_t}$$

where

$$\begin{aligned} \tfrac{1}{2}\delta_2(t) b(t,T)^2 + \hat{\kappa}(t) b(t,T) - \tfrac{\partial b(t,T)}{\partial t} - 1 &= 0 \\ \tfrac{\partial a(t,T)}{\partial t} + \hat{\varphi}(t) b(t,T) - \tfrac{1}{2}\delta_1(t) b(t,T)^2 &= 0 \end{aligned}$$

with terminal conditions $a(T,T) = b(T,T) = 0$. It is proven by showing that the pricing function

$$\frac{\partial B^T(r,t)}{\partial t} + \hat{\alpha}\left(r_t, t\right) \frac{\partial B^T(r,t)}{\partial r} + \frac{1}{2}\hat{\beta}\left(r_t, t\right)^2 \frac{\partial^2 B^T(r,t)}{\partial r^2} - r B^T(r,t) = 0$$

along with $B^T(r,T) = 1$ is satisfied (look at terminal condition, find derivatives and plug in and observe that it is satisfied for the above equations).

We now have that $a(\cdot)$ and $b(\cdot)$ depend on both calender time t and maturity T. The above equations can be hard to solve of all parameters depend on t. If we only want to match the current yield curve it is enough to let $\hat{\varphi}(t)$ be time dependent and leave the other constant. With this, the yield curve volatility will only depend on time to maturity (reasonable, since volatility is approximately stable). The τ-maturity yield satisfies

$$y_t^{t+\tau} = \frac{a(t, t+\tau)}{\tau} + \frac{b(t, t+\tau)}{\tau} r_t$$

Hence, its dynamics have the form (Itô)

$$\mathrm{d}y_t^{t+\tau} = [\ldots]\ \mathrm{d}t + \frac{b(t, t+\tau)}{\tau} \beta\left(r_t, t\right)\ \mathrm{d}z_t$$

The price of the τ-maturity zero-coupon bond is

$$B_t^{t+\tau} = \mathrm{e}^{-a(t,t+\tau) - b(t,t+\tau) r_t}$$

with dynamics (Itô)

$$\mathrm{d}B_t^{t+\tau} = [\ldots]\ \mathrm{d}t - B_t^{t+\tau} b(t, t+\tau) \beta\left(r_t, t\right)\ \mathrm{d}z_t$$

Ideally we want $b(t, t+\tau)\beta\left(r_t, t\right)$ not to depend on t. By choosing appropriate parameters as constants $(\delta_1, \delta_2, \hat{\kappa})$ we solve this. Obvious for β and the ODE of $b(t,T)$ becomes

$$\frac{1}{2}\delta_2 b(t,T)^2 + \hat{\kappa} b(t,T) - \frac{\partial b(t,T)}{\partial t} - 1 = 0$$

This is the same as for the time-homogeneous case $(b(t,T) = b(T-t))$. So, by choosing appropriate parameters as constant $y_t^{t+\tau}$ and $B_t^{t+\tau}$ only depend on τ and not calender time t.

Given time dependent $\hat{\varphi}(t)$, let's solve the ODE of $a(t,T)$. Using $a(T,T) = 0$, integration gives $(b(t,T) = b(T-t))$

$$a(t,T) = \int_t^T \hat{\varphi}(t) b(T-u)\ \mathrm{d}u - \frac{1}{2}\delta_1 \int_t^T b(T-u)^2\ \mathrm{d}u$$

We denote the current (time 0) price observed in the market of a T-bond with $\bar{B}(T)$ (i.e. $B_0^T = \bar{B}(T)$). To match the market price, for any T we require

$$\bar{B}(T) = \exp\left(-a(0,T) - b(T)r_0\right) \Longleftrightarrow a(0,T) = -\ln(\bar{B}(T)) - b(T)r_0$$

(Compare with $a(t,T)$ above with $t = 0$). For some models the functions $\hat{\varphi}(t)$ and $a(t,T)$ can be computed in closed form. For other, numerical approximations are needed.

We have managed to match the current yield curve. To obtain a model that is consistent with the current volatility of the yield curve and prices of liquid bonds and derivatives (caps, floors, etc.), we have to pay the price of having 2 or 3 time dependent parameters. (Extended Vasicek, 2 time dependent parameters can match current yield curve and current yield volatilities.)

Additional stuff

Ho-Lee model: (Extended Merton), the $\mathbb{Q}$-dynamics are

$$\mathrm{d}r_t = \hat{\varphi}(t)\ \mathrm{d}t + \beta\ \mathrm{d}z_t^{\mathbb{Q}}$$

Bond prices are

$$B^T\left(r_t, t\right) = \mathrm{e}^{-a(t,T) - b(T-t)r_t}$$

with

$$b(\tau) = \tau, \quad a(t,T) = \int_t^T \hat{\varphi}(u)(T-u)\ \mathrm{d}u - \frac{1}{2}\beta^2 \int_t^T (T-u)^2\ \mathrm{d}u$$

Let $t \mapsto \bar{f}(t)$ be the current term structure of forward rates and be diffrentiable. Moreover, suppose that for all t

$$\hat{\varphi}(t) = \frac{\partial \bar{f}(t)}{\partial t} + \beta^2 t$$

Then the Ho-Lee model will reproduce the current term structure, and we obtain

$$a(t,T) = -\ln\left(\frac{\bar{B}(T)}{\bar{B}(t)}\right) - (T-t)\bar{f}(t) + \frac{1}{2}\beta^2 t(T-t)^2$$

Extended Vasicek model: (Hull-White), the $\mathbb{Q}$-dynamics are

$$\mathrm{d}r_t = \left(\hat{\varphi}(t) - \kappa r_t\right)\ \mathrm{d}t + \beta\ \mathrm{d}z_t^{\mathbb{Q}}$$

with bond price

$$B^T\left(r_t, t\right) = e^{-a(t,T) - b(T-t)r_t}.$$

Here ($b(\tau)$ as in the Varsicek model)

$$b(\tau) = \frac{1 - e^{-\kappa\tau}}{\kappa}, \quad a(t,T) = \int_t^T \hat{\phi}(u) b(T-u)\ \mathrm{d}u + \frac{1}{2}\beta^2 \int_t^T b(T-u)^2\ \mathrm{d}u.$$

Let $t \mapsto \overline{f}(t)$ be the current term structure of forward rates and be differentiable. Then

$$\hat{\phi}(t) = \overline{f}(t) + \frac{1}{\kappa}\frac{\partial \overline{f}(t)}{\partial t} + \frac{\beta^2}{1\kappa^2}(1 - e^{-2\kappa t}),$$

will math the current term structure and we have:

$$a(t,T) = -\ln\left(\frac{\overline{B}(T)}{\overline{B}(t)}\right) - b(T-t)\overline{f}(t) + \frac{\beta^2}{1\kappa} b(T-t)^2 (1 - e^{-2\kappa t}).$$

8 Forward rate models

Classical short rate models do not fit the currently observed yield curve. We have solved this with time-inhomogeneous models. However, it is not (from an economic point of view) reasonable that the entire yield curve is driven by the short rate. Moreover, it is hard in a short rate model to obtain a realistic volatility structure for forward rates while retaining tractability of the model. A more natural approach for fitting the yield curve is to start from the observed yield curve and then model the dynamics of the entire curve. The Heath-Jarrow-Morton (HJM) approach models the instantaneous forward rates.

- An approach to model the entire curve of forward rates.
- The HJM restriction pins down the drift of the forward rates.
- The volatility of forward rates should be bounded from above.

Forward rates

The continuously compounded forward rate prevailing at time t for the period $[T, T+\delta]$ is

$$f_t^{T,T+\delta} = -\frac{\ln\left(B_t^{T+\delta}\right) - \ln\left(B_t^T\right)}{\delta}.$$

The instantaneous T-maturity forward rate at time t is

$$f_t^T = \lim_{\delta\to 0} f_t^{T,T+\delta} = -\frac{\partial \ln\left(B_t^T\right)}{\partial T} = y_t^T + (T-t)\frac{\partial y_t^T}{\partial T}.$$

(The forward rate curve f_t^T consists of infinitely many rates).
Suppose the current forward rates $T \mapsto f_0^T$ are observed. For each T, we model the dynamics of f_t^T as

$$\mathrm{d}f_t^T = \alpha\left(t, T, (f_t^s)_{s\geq t}\right)\,\mathrm{d}t + \sum_{i=1}^n \beta_i\left(t, T, (f_t^s)_{s\geq t}\right)\,\mathrm{d}z_{it}, \quad 0 \leq t \leq T$$

where $z_1, \ldots, z_n$ are independent $\mathbb{P}$-BMs.

- The current yield curve is matched since we observe $T \mapsto f_0^T$.
- We have 1 SDE for each T all driven by the same n random shocks.
- Drift/volatility may depend on the entire forward rate curve $(f_t^s)_{s\geq t}$ prevailing at time t.
- Hence, infinitely many state variables (PDE approach not suitable), so we use martingale pricing.
- Typically, drift/volatility terms have simpler forms ($\alpha(t,T)$ or $\alpha\left(t, T, f_t^T\right)$).
  $-r_t = f_t^t$

Zero-coupon bond price

The price B_t^T under $\mathbb{P}$ of a zero-coupon bond price maturing at time T satisfies

$$\mathrm{d}B_t^T = B_t^T\left(\mu^T\left(t, (f_t^s)_{s\geq t}\right)\,\mathrm{d}t + \sum_{i=1}^n \sigma_i^T\left(t, (f_t^s)_{s\geq t}\right)\,\mathrm{d}z_{it}\right)$$

where

$$\mu^T\left(t, (f_t^s)_{s\geq t}\right) = r_t - \int_t^T \alpha\left(t, u, (f_t^s)_{s\geq t}\right)\,\mathrm{d}u + \frac{1}{2}\sum_{i=1}^n\left(\sigma_i^T\left(t, (f_t^s)_{s\geq t}\right)\right)^2$$

$$\sigma_i^T\left(t, (f_t^s)_{s\geq t}\right) = -\int_t^T \beta_i\left(t, u, (f_t^s)_{s\geq t}\right)\,\mathrm{d}u$$

Proof. Consider the case of $n=1$. Then (suppress dependency on $(f_t^s)_{s\geq t}$)

$$\begin{aligned} \mathrm{d}f_t^T &= \alpha\left(t,T,(f_t^s)_{s\geq t}\right)\,\mathrm{d}t + \beta\left(t,T,(f_t^s)_{s\geq t}\right)\,\mathrm{d}z_t \\ &= \alpha(t,T)\,\mathrm{d}t + \beta(t,T)\,\mathrm{d}z_t \end{aligned}$$

Denote

$$Y_t = \int_t^T f_t^s\,\mathrm{d}s$$

Using Leibnitz' rule on Y_t we have

$$\mathrm{d}Y_t = \left(-r_t + \int_t^T \alpha(t,u)\,\mathrm{d}u\right)\mathrm{d}t + \left(\int_t^T \beta(t,u)\,\mathrm{d}u\right)\mathrm{d}z_t$$

By using Itô's Lemma on

$$B_t^T = \mathrm{e}^{-\int_t^T f_t^s\,\mathrm{d}s} = \mathrm{e}^{-Y_t} = g(Y_t), \quad \left(g'(Y_t) = -\mathrm{e}^{-Y_t}, \quad g''(Y_t) = \mathrm{e}^{-Y_t}\right)$$

we find

$$\begin{aligned} \mathrm{d}B_t^T &= -\mathrm{e}^{-Y_t}\,\mathrm{d}Y_t + \frac{1}{2}\mathrm{e}^{-Y_t}(\,\mathrm{d}Y_t)^2 \\ &= -\mathrm{e}^{-Y_t}\left(\left(-r_t + \int_t^T \alpha(t,u)\,\mathrm{d}u\right)\mathrm{d}t + \left(\int_t^T \beta(t,u)\,\mathrm{d}u\right)\mathrm{d}z_t\right) + \frac{1}{2}\mathrm{e}^{-Y_t}\left(\int_t^T \beta(t,u)\,\mathrm{d}u\right)^2 dt \\ &= B_t^T\left(r_t - \int_t^T \alpha(t,u)\,\mathrm{d}u + \frac{1}{2}\left(\int_t^T \beta(t,u)\,\mathrm{d}u\right)^2\right)\mathrm{d}t - B_t^T\left(\int_t^T \beta(t,u)\,\mathrm{d}u\right)\mathrm{d}z_t \end{aligned}$$

HJM drift restriction

Proceeding as in the proof above, we find that the drift rate of the zero-coupon bond under $\mathbb{Q}$ is

$$r_t - \int_t^T \hat{\alpha}\left(t,u,(f_t^s)_{s\geq t}\right)\mathrm{d}u + \frac{1}{2}\sum_{i=1}^n\left(\int_t^T \beta_i\left(t,u,(f_t^s)_{s\geq t}\right)\mathrm{d}u\right)^2$$

Meaning the drift must equal r_t in the absence of arbitrage. Therefore

$$\int_t^T \hat{\alpha}\left(t,u,(f_t^s)_{s\geq t}\right)\mathrm{d}u = \frac{1}{2}\sum_{i=1}^n\left(\int_t^T \beta_i\left(t,u,(f_t^s)_{s\geq t}\right)\mathrm{d}u\right)^2$$

Differentiating w.r.t. T gives the HJM drift restriction

$$\hat{\alpha}\left(t,T,(f_t^s)_{s\geq t}\right) = \sum_{i=1}^n \beta_i\left(t,T,(f_t^s)_{s\geq t}\right)\int_t^T \beta_i\left(t,u,(f_t^s)_{s\geq t}\right)\mathrm{d}u$$

This is a beautiful result, because as β_i can be estimated by data so can $\hat{\alpha}$.

Additional stuff

Leibnitz' rule: For any $s\in[1,T]$, let the dynamics of $(f_t^s)_{s\in[0,s]}$ be given by

$$\mathrm{d}f_t^s = \alpha_t^s\,\mathrm{d}t + \beta_t^s\,\mathrm{d}z_t$$

Let's define $Y_t = \int_t^T f_t^s\,\mathrm{d}s$. Then

$$\mathrm{d}Y_t = \left(\int_t^T \alpha_t^s\,\mathrm{d}s - f_t^t\right)\mathrm{d}t + \left(\int_t^T \beta_t^s\,\mathrm{d}s\right)\mathrm{d}z_t$$

Forward rate dynamics under $\mathbb{Q}$: Given the market price of risk $\lambda = (\lambda_1, \ldots, \lambda_n)^T$, Girsanov's Theorem states that $\mathrm{d}z_{it}^{\mathbb{Q}} + \lambda_{it}\ \mathrm{d}t$ is a Wiener process under $\mathbb{Q}$, hence

$$\mathrm{d}f_t^T = \hat{\alpha}\left(t, T, (f_t^s)_{s\geq t}\right)\ \mathrm{d}t + \sum_{i=1}^{n} \beta_i\left(t, T, (f_t^s)_{s\geq t}\right)\ \mathrm{d}z_{it}^{\mathbb{Q}}, \quad 0 \leq t \leq T,$$

where

$$\hat{\alpha}\left(t, T, (f_t^s)_{s\geq t}\right) = \alpha\left(t, T, (f_t^s)_{s\geq t}\right) - \sum_{i=1}^{n} \beta_i\left(t, T, (f_t^s)_{s\geq t}\right)\lambda_{it}$$

Procedure for using HJM models: (n = 1 for simplicity). Pick some $\beta(t, T)$ (short form of $\beta\left(t, u\, (f_t^s)_{s\geq t}\right)$). The drift is given by

$$\hat{\alpha}(t, T) = \beta(t, T) \int_t^T \beta(t, u)\ \mathrm{d}u$$

Observe current forward rate $T \mapsto f(0, T)$ from the market. Then, forward rates are given by

$$f(t, T) = f(0, T) + \int_0^t \hat{\alpha}(u, T)\ \mathrm{d}u + \int_0^T \beta(u, T)\ \mathrm{d}z_u^{\mathbb{Q}}$$

We can find bond prices by $B_t^T = \mathrm{e}^{-\int_t^T f_t^s\ \mathrm{d}s}$. Given B_t^T and f_t^T we may find other derivative prices. Note, we do not need to specify drift.

The Ho-Lee model: (extended Merton). A one-factor model with $\beta\left(t, T, (f_t^s)_{s\geq t}\right) = \beta > 0$. The HJM drift restriction is

$$\hat{\alpha}\left(t, T, (f_t^s)_{s\geq t}\right) = \beta^2(T - t).$$

T-forward rates

$$f_t^T = f_0^T + \int_0^t \beta^2(T - u)\ \mathrm{d}u + \int_0^t \beta\ \mathrm{d}z_u^{\mathbb{Q}}$$

For $T = t$ we have

$$r_t = f_t^t = f_0^t + \frac{1}{2}\beta^2 t^2 + \int_0^t \beta\ \mathrm{d}z_u^{\mathbb{Q}}, \quad \Longrightarrow \quad \mathrm{d}r_t = \hat{\varphi}(t)\ \mathrm{d}t + \beta\ \mathrm{d}z_t^{\mathbb{Q}}$$

with $\hat{\varphi}(t) = \frac{\partial f_0^t}{\partial t} + \beta^2 t$.
Market models: The entire forward rate curve is the modeling object in the HJM approach. However, instantaneous forward rates are not observed in practice. This lead
researchers to propose market models that rely on observed forward rates. They use periodically compounded interest rates and take the currently observed yield curve as given. It is possible to get Black-Scholes type formulas for caps, floors and swaptions. No model gives exact closed-form formulas for the prices of caps/floors and swaptions simultaneously.

9 Structural models of default

One framework to model credit risk (default risk) is structural models (most intuitive). Here, the value of the firm is modeled explicitly. Useful for corporations to determine corporate finance related decisions, e.g. optimal debt capital structure, decision to default, etc.

One has to use models when no replicating portfolio exists, the replicating portfolio consists of illiquid instruments (prices are unreliable), or in the case of multi-name derivatives the transaction costs explode when setting up the replicating portfolio (CDO with more than 100 underlying names).

Black-Scholes-Merton model

Assume a firm is financed by one zero-coupon bond with face value F and maturity T (the present value of this liability equals $\tilde{B}_t$) and one equity share with time t value S_t. The basic idea of the BSM model is to define default as the situation when the asset value of the firm is less than the face value of the debt at maturity of the debt, $V_t < F$. Thus, time of deafult is

$$\tau = \begin{cases} T, & \text{if } V_T < F \\ \infty, & \text{if } V_T \geq F \end{cases}$$

The payouts associated with debt and equity have the following representation

$$\begin{aligned} \tilde{B}_T &= \min\{F, V_T\} = F - \max\{F - V_T, 0\} \\ S_T &= \max\{V_T - F, 0\} \end{aligned}$$

The valuation of these reduce to valuation of European put and call options

$$\begin{aligned} \tilde{B}_t &= \mathrm{e}^{r(T-t)}F - p(t, V_t; T, F) \\ &= \mathrm{e}^{r(T-t)}F - \left(c(t, V_t; T, F) - V_t + \mathrm{e}^{r(T-t)}F\right) \\ &= V_t - c(t, V_t; T, F) \\ S_t &= c(t, V_t; T, F) = V_t - \mathrm{e}^{r(T-t)}F + p(t, V_t; T, F), \end{aligned}$$

using the put-call parity. To find these prices we need to impose some assumptions and dynamics on the underlying asset value process V_t.

Model assumptions: No transaction costs or taxes. Trading takes place continuously in time. The borrowing and lending interest rate are the same and constant r, i.e. $R(t,T) = r$ for all t and T. No restrictions on short selling. No arbitrage opportunities in the market. The firm value V_t is a traded asset and follows a geometric BM

$$\mathrm{d}V_t = \mu V_t \, \mathrm{d}t + \sigma V_t \, \mathrm{d}W_t$$

where μ is the drift, σ the volatility of the Wiener process.
To arrive at the risk neutral dynamics we just need to adjust the drift so it is equal to the interest rate r

$$\mathrm{d}V_t = r V_t \, \mathrm{d}t + \sigma V_t \, \mathrm{d}W_t$$

We know that the asset value is log-normal, so the risk neutral default probability is

$$
\begin{aligned}
\mathbb{Q}[\tau < \infty] &= \mathbb{Q}[V_T < F] \\
&= \mathbb{Q}\left[V_0 e^{\left(r-\frac{1}{2}\sigma^2\right)T+\sigma W_T} < F\right] \\
&= \mathbb{Q}\left[\left(r-\frac{1}{2}\sigma^2\right)T + \sigma W_T < \ln\left(\frac{F}{V_0}\right)\right] \\
&= \mathbb{Q}\left[W_T < \frac{\ln\left(\frac{F}{V_0}\right) - \left(r-\frac{1}{2}\sigma^2\right)T}{\sigma}\right] \\
&= \mathbb{Q}\left[W_1 < \frac{\ln\left(\frac{F}{V_0}\right) - \left(r-\frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}\right] \\
&= \Phi\left(\frac{\ln\left(\frac{F}{V_0}\right) - \left(r-\frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}\right),
\end{aligned}
$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function.

$$
\mathbb{Q}_t[\tau < \infty] = \Phi\left(\frac{\ln\left(\frac{F}{V_t}\right) - \left(r-\frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}\right)
$$

We have the well known Black-Scholes formula for the price of a call option, so the debt and equity values are

$$
\begin{aligned}
\tilde{B}_t &= V_t - c(t, V_t; T, F) = V_t - \left(V_t\Phi(d_1) - e^{-r(T-t)}F\Phi(d_2)\right) \\
S_t &= c(t, V_t; T, F) = V_t\Phi(d_1) - e^{-r(T-t)}F\Phi(d_2)
\end{aligned}
$$

where

$$
\begin{aligned}
d_1 &= \frac{1}{\sigma\sqrt{T-t}}\left(\ln\left(\frac{V_t}{F}\right) + \left(r+\frac{1}{2}\sigma^2\right)(T-t)\right) \\
d_2 &= d_1 - \sigma\sqrt{T-t}
\end{aligned}
$$

Additional stuff

Credit spread: The credit spread on a corporate bond is the difference between the yield to maturity of the bond deducted the yield on a bond with the same characteristics. Thus, the credit spread in the BSM model is

$$
\begin{aligned}
\zeta_t^T &= \tilde{y}_t^T - y_t^T \\
&= \frac{1}{T-t}\ln\left(\frac{F}{\tilde{B}_t^T}\right) - r \\
&= -\frac{1}{T-t}\ln\left(\frac{1}{l_t}\Phi(-d_1) + \Phi(d_2)\right)
\end{aligned}
$$

where $l_t = \frac{e^{-r(T-t)}F}{V_t}$ is the firms leverage ratio. Furthermore, $\frac{dS(t,T)}{dl_t} > 0$, meaning the spread is increasing in the leverage ratio (unrealistic?).

Black-Cox model: A critique point of the BSM model is that a company cannot default before maturity. In The Black-Cox model a company defaults if the value of the assets falls below some exogenously given barrier.

10 Reduced form models of default

A framework for modeling credit risk. Less intuitive than structural models, but easier to implement. Here, the default probability is modeled by the default intensity process. Useful for modeling of derivatives. Flexible approach.

One has to use models when no replicating portfolio exists, the replicating portfolio consists of illiquid instruments (prices are unreliable), or in the case of multi-name derivatives the transaction costs explode when setting up the replicating portfolio (CDO with more than 100 underlying names).

With reduced form models we model the default of a firm directly. (For structural models we need to observe a lot of information for each firm, all data rarely avaible, very complex (hard to compute derivatives prices), unrealistic short- and long-term spreads). We do not try to tie the fundamentals of the firm to the random default time of a company. Instead the time of default τ is modeled directly and not in relation to the balance sheet. τ is simply modeled as the first jump time of some underlying process N_t, hence (τ is a stopping time)

$$\tau = \inf\{t > 0 \mid N_t > 0\}.$$

The approach if the specify N_t (or how it jumps) so τ has the desired properties.
We call the information generated by the background process X_t the background filtration

$$\{\mathcal{G}_t\}_{t\geq 0} = \{\sigma\{X_s \mid s \leq t\}\}_{t\geq 0}$$

All the non-defaultable processes and the intensity process λ_t is adapted to $\{\mathcal{G}_t\}_{t\geq 0}$. The full filtration $\mathcal{F}_t$ is generated by the background filtration and the filtration generated by N_t

$$\{\mathcal{F}_t\}_{t\geq 0} = \{\mathcal{G}_t\}_{t\geq 0} \vee \{\sigma\{N_s \mid s \leq t\}\}_{t\geq 0}$$

In the intensity reduced form models, the modeling object is the intensity process λ_t by which the default process N_t jumps. We have looked at three different choices Constant intensity λ, deterministic intensity function λ_t, and stochastic intensity process driven by $X_t, \lambda(X_t)$.

10.1 Stochastic intensity reduced form model

Letting the intensity be stochastic results in the default process N_t being a Cox process. In this setup the survival probabilities are given by

$$\mathbb{Q}_t[\tau > T] = \mathbb{E}_t\left[e^{-\int_t^T \lambda_s \, ds}\right]$$

This is completely analogue to pricing of risk-free ZCBs, so all the short interest rate modeling machinery can be used.

CIR intensity: Letting the default intensity follow a CIR process

$$d\lambda_t = k(\bar{\lambda} - \lambda)\, dt + \sigma\sqrt{\lambda_t}\, dW_t,$$

we have a tractable specification. The intensity varies ramdomly with a tendency to revert to its long-run mean $\bar{\lambda}$ (k determines the speed of reversion). With the Feller conditions

$$2k\bar{\lambda} > \sigma^2, \quad \text{and} \quad \lambda_0 > 0,$$

it can be shown that λ_t is strictly positive for all t. In this setting the survival probability is

$$\mathbb{Q}_t[\tau > T] = e^{a(T-t) - b(T-t)\lambda_t}$$

where a and b are deterministic functions of time to maturity with

$$a(T-t) = \frac{2k\bar{\lambda}}{\sigma^2}\ln\left(\frac{2\gamma e^{\frac{1}{2}(\gamma+k)(T-t)}}{(\gamma+k)\left(e^{\gamma(T-t)} - 1\right) + 2\gamma}\right)$$

$$b(T-t) = \frac{2\left(e^{\gamma(T-t)} - 1\right)}{(\gamma+k)\left(e^{\gamma(T-t)} - 1\right) + 2\gamma}$$

where $\gamma = \sqrt{k^2 + 2\sigma^2}$.

10.2 Pricing defaultable bonds (zero recovery)

With stochastic intensities and interest rates it can be shown that with $R = 0$

$$\begin{aligned}
\tilde{B}_t^T &= \mathbb{E}_t\left[e^{-\int_t^T r_s \, ds} \mathbb{1}_{\{\tau > T\}}\right] \\
&= \mathbb{E}_t\left[\mathbb{E}\left[e^{-\int_t^T r_s \, ds} \mathbb{1}_{\{\tau > T\}} \mid \mathcal{G}_T\right]\right] \\
&= \mathbb{E}_t\left[\mathbb{E}\left[e^{-\int_t^T r_s \, ds} \mathbb{1}_{\{N_T = 0\}} \mid \mathcal{G}_T\right]\right] \\
&= \mathbb{E}_t\left[\mathbb{E}\left[e^{-\int_t^T r_s \, ds} \mid \mathcal{G}_T\right] \mathbb{E}\left[\mathbb{1}_{\{N_T = 0\}} \mid \mathcal{G}_T\right]\right] \\
&= \mathbb{E}_t\left[\mathbb{E}\left[e^{-\int_t^T r_s \, ds} \mid \mathcal{G}_T\right] e^{-\int_t^T \lambda_s \, ds}\right] \\
&= \mathbb{E}_t\left[e^{-\int_t^T r_s + \lambda_s \, ds}\right]
\end{aligned}$$

If the intensity process λ_t and interest rate process r_t are independent

$$\tilde{B}_t^T = \mathbb{E}_t\left[e^{-\int_t^T r_s \, ds}\right] \mathbb{E}_t\left[e^{-\int_t^T \lambda_s \, ds}\right] = B_t^T \mathbb{Q}_t[\tau > T]$$

as in the case with deterministic intensity.

Additional stuff

Possion process: Let N_t be a poisson process with intensity parameter λ (this means $N_t \sim po(\lambda t)$). Then

$$\mathbb{Q}[N_t = k] = \frac{(\lambda t)^k}{k!} e^{-\lambda t}$$

11 Credit derivatives

The main idea when pricing a credit default swap (CDS) (a single name credit derivative) is that the value at time $t = 0$ is zero. We set spreads so we are indifferent between being a protection seller or buyer (in expectation).

The protection buyer pays the CDS spread $s^{CDS}(T)$ every payment date unless the reference entity goes bankrupt. The value of the stream of cashflows is the sum of the discounted future payments times the probability of no default until time t. (Assuming the CDS does not carry any counterparty risk itself).

Pricing CDS

Let $\delta_i = T_i - T_{i-1}$. The protections buyer's cashflows are

$$\begin{aligned}\pi^{\text{prem}} &= \mathbb{E}^{\mathbb{Q}}\left[\sum_{i=1}^{N} \exp\left(-\int_0^{T_i} r_s \, \mathrm{d}s\right)\mathbb{1}_{\{\tau > T_i\}} s^{CDS}(T_N)\delta_i\right] \\ &= s^{CDS}(T_N)\sum_{i=1}^{N}\delta_i \underbrace{\mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\int_0^{T_i} r_s + \lambda_s \, \mathrm{d}s\right)\right]}_{\text{price of defaultable ZCB, no recovery}} \\ &= s^{CDS}(T_N)\sum_{i=1}^{N}\delta_i B_0^{T_i}\mathbb{Q}_0[\tau > T_i]\end{aligned}$$

The protection seller pays $(1-R)\cdot$ Notional if there is a default. (We abstract from settlement, liquidity and other issues). Assuming recovery rate R constant, the protection seller's cashflows are

$$\begin{aligned}\pi^{\text{prot}} &= \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\int_0^{\tau} r_s \, \mathrm{d}s\right)\mathbb{1}_{\{\tau \leq T_N\}}(1-R)\right] \\ &= (1-R)\mathbb{E}^{\mathbb{Q}}\left[\int_0^{T_N} \lambda_t \exp\left(-\int_0^{t} r_s + \lambda_s \, \mathrm{d}s\right) \mathrm{d}t\right]\end{aligned}$$

If r and λ are independent

$$\pi^{\text{prot}} = (1-R)\int_0^{T_N} B_o^t \mathbb{Q}_0[\tau > t]\hat{\lambda}_t \, \mathrm{d}t$$

with $\hat{\lambda}(t)$ being the time-deterministic intensity (or expected intensity $\mathbb{E}_0^{\mathbb{Q}}[\lambda_t]$).
Equating the two $\pi^{\text{prem}} = \pi^{\text{prot}}$ we obtain

$$s^{CDS}(T_N)\sum_{i=1}^{N}\delta_i B_0^{T_i}\mathbb{Q}_0[\tau > T_i] = (1-R)\int_0^{T_N} B_o^t \mathbb{Q}_0[\tau > t]\hat{\lambda}_t \, \mathrm{d}t$$

$$\Updownarrow$$

$$s^{CDS}(T_N) = \frac{(1-R)\int_0^{T_N} B_o^t \mathbb{Q}_0[\tau > t]\hat{\lambda}_t \, \mathrm{d}t}{\sum_{i=1}^{N}\delta_i B_0^{T_i}\mathbb{Q}_0[\tau > T_i]}$$

If λ is constant $\mathbb{Q}_0[\tau > T_i] = \mathrm{e}^{-\lambda T_i}$. If λ is affine

$$\mathbb{Q}_0[\tau > T_i] = \mathbb{E}^{\mathbb{Q}}\left[\mathrm{e}^{-\int_0^{T_i}\lambda_s \, \mathrm{d}s}\right] = \mathrm{e}^{A(0,T_i)+B(0,T_i)\lambda_0}$$

Marking to market

To find the current value (time $t, t_{n-1} \leq t < t_n$) of the contract (an old one) from the perspective of the protection buyer, observe that

$$V_{\text{old}} = V_{\text{old}} + V_{new},$$

since the newly issued contract has zero market value. Thus

$$
\begin{aligned}
V_{\text{old}} &= \pi_{\text{old}}^{\text{prot}} - \pi_{\text{old}}^{\text{prem}} - \left(\pi_{\text{new}}^{\text{prot}} - \pi_{\text{new}}^{\text{prem}}\right) \\
&= \pi_{\text{new}}^{\text{prem}} - \pi_{\text{old}}^{\text{prem}} \\
&= \sum_{i=1}^{N} B_t^{T_i} \mathbb{Q}_t[\tau > T_i] \delta_i s_t^{CDS}(T_N) - \sum_{i=1}^{N} B_t^{T_i} \mathbb{Q}_t[\tau > T_i] \delta_i s_0^{CDS}(T_N) \\
&= \sum_{i=1}^{N} B_t^{T_i} \mathbb{Q}_t[\tau > T_i] \delta_i \left(s_t^{CDS}(T_N) - s_0^{CDS}(T_N)\right),
\end{aligned}
$$

since $\pi_{\text{old}}^{\text{prot}} = \pi_{\text{new}}^{\text{prot}}$. From this it can be seen that the value of the contract from the perspective of the protection buyer increases as the CDS spread increases.

Additional stuff

Calibration: A short how to - We have market quotes $X_t = (x_0, x_1, \dots, x_n)$ and closed form solution $\pi_X = f(\theta)$. What value of θ should we choose so $\pi_X(i)$ is close to x_i for all $i = 0, 1, \dots, n$?

11.1 Multi-name models of default

We consider credit derivatives whose underlying is a set of names (reference entities). A multi-name contract is associated with multiple credit events which trigger remedial payment to the protection buyer. The collateralized debt obligation (CDO) is one of the most popular multi-name credit derivatives. We want to look at the default correlation.

A structual approach

To model the default correlation consider the Black-Scholes-Merton (BSM) structural model of default with default defined as

$$\tau = \begin{cases} T, & \text{if } V_T < F \\ \infty & \text{if } V_T \geq F \end{cases}$$

where F is the face value of the debt and V_T is the asset values at maturity. Hence, it is the asset values V_T at maturity T that is the quantity of interest. Assuming $T = 1$ and defining the return on the asset values $\tilde{R}$ over the period from 0 to T as

$$e^{\tilde{R}} = \frac{V_1}{V_0} = \frac{V_0 e^{\left(\mu - \frac{\sigma^2}{2}\right)}}{V_0}$$

where the last equality holds since V_T is log-normally distributed in the BSM model. Taking the logarithm on both sides we have

$$\tilde{R} \sim N\left(\mu - \frac{\sigma^2}{2}, \sigma^2\right)$$

Likewise, the default can be written in terms of the return on the asset value

$$\begin{gathered} V_1 < F, \\ \widehat{\Downarrow} \\ \tilde{R} = \ln\left(\frac{V_1}{V_0}\right) < \ln\left(\frac{F}{V_0}\right), \\ \widehat{\Downarrow} \\ R = \frac{\tilde{R} - \left(\mu - \frac{\sigma^2}{2}\right)}{\sigma} < \frac{\ln\left(\frac{F}{V_0}\right) - \left(\mu - \frac{\sigma^2}{2}\right)}{\sigma} = C, \end{gathered}$$

where R is the standardized return on the firm value and the right hand side is a constant C. Thus, default of the firm can be written as

$$R < C$$

where R is standard normally distributed.

11.2 Introducing several companies

We define the default of company i as when the standardized return on owning the firm falls below a firm specific threshold C_i at some future time T

$$\text{Company } i \text{ defaults at time } T \Longleftrightarrow R_i \leq C_i, \quad R_i = \frac{\tilde{R}_i - \mathbb{E}\left[\tilde{R}_i\right]}{\sqrt{\mathbb{V}\left[\tilde{R}_i\right]}}$$

The standardization means all R_i s in the portfolio have zero mean and unit variance. We assume that individual period returns R_i are normally distributed. Thus, the default probabilities are

$$q_i = \mathbb{Q}\left[R_i \leq C_i\right] = \Phi\left(C_i\right) \quad \Longrightarrow \quad C_i = \Phi^{-1}\left(q_i\right)$$

with $\Phi(\cdot)$ the CDF of the standard normal distribution. Furthermore, assume that all returns are correlated through one common factor α by

$$R_i = \beta_i \alpha + \sqrt{1-\beta_i^2}\,\varepsilon_i$$

where α and ε_i are independent and standard normally distributed. Also, the ε_i s are independent. (The common factor α could be interpreted as the standardized return on the stock market or the general state of the economy).

How are company returns correlated in this setting?

$$\begin{aligned}
\text{Cov}\left(R_i, R_j\right) &= \mathbb{E}\left[(R_i - \mathbb{E}[R_i])\left(R_j - \mathbb{E}\left[R_j\right]\right)\right] \\
&= \mathbb{E}\left[R_i R_j\right] \\
&= \mathbb{E}\left[\beta_i \beta_j \alpha^2\right] \\
&= \beta_i \beta_j \\
&= \text{corr}\left(R_i, R_j\right),
\end{aligned}$$

since all the other random variables involved are independent and we have unit variance. Key advantage: Given a common factor α the returns are independent, observed by

$$\begin{aligned}
&\mathbb{E}[R_i \mid \alpha] = \beta_i \alpha \\
&\mathbb{V}[R_i \mid \alpha] = \mathbb{E}\left[(R_i - \beta_i \alpha)^2 \mid \alpha\right] = \mathbb{E}\left[\left(1-\beta_i^2\right)\varepsilon_i^2 \mid \alpha\right] = 1 - \beta_i^2
\end{aligned}$$

Now, we can find the correlation of the returns given knowledge of the common factor α

$$\text{Cov}\left(R_i, R_j \mid \alpha\right) = \sqrt{1-\beta_i^2}\sqrt{1-\beta_j^2}\,\mathbb{E}\left[\varepsilon_i \varepsilon_j \mid \alpha\right] = 0$$

since individual factors were assumed independent. From this, the convenient form of the default probability given α arises

$$\begin{aligned}
q_i(\alpha) &= \mathbb{Q}[R_i \le C_i \mid \alpha] \\
&= \mathbb{Q}\left[\varepsilon_i \le \frac{C_i - \beta_i \alpha}{\sqrt{1-\beta_i^2}} \,\middle|\, \alpha\right] \\
&= \Phi\left(\frac{C_i - \beta_i \alpha}{\sqrt{1-\beta_i^2}}\right) \\
&= \Phi\left(\frac{\Phi^{-1}(q_i) - \beta_i \alpha}{\sqrt{1-\beta_i^2}}\right)
\end{aligned}$$

which depends on β_i.

What are the correlation of defaults $\rho_{i,j}$? It has the form form

$$\rho_{i,j} = \frac{q_{i\&j} - q_i q_j}{\sqrt{q_i(1-q_i)}\sqrt{q_j\left(1-q_j\right)}} = \frac{\Phi_2\left(C_i, C_j; \beta_i \beta_j\right) - \Phi(C_i)\Phi\left(C_j\right)}{\sqrt{\Phi(C_i)(1-\Phi(C_i))}\sqrt{\Phi\left(C_j\right)\left(1-\Phi\left(C_j\right)\right)}},$$

using $q_i(\alpha) = \Phi(C_i)$ and that R_i and R_j are bivariate normally distributed with correlation coefficient $\text{corr}\left(R_i, R_j\right) = \beta_i \beta_j$ so

$$q_{i\&j} = \mathbb{Q}\left[R_i \le C_i \text{ and } R_j \le C_j\right] = \Phi_2\left(C_i, C_j; \beta_i \beta_j\right)$$

Additional stuff

The loss distribution function: Most multi-name credit derivatives depend on the losses incurred in the portfolio. Hence, the object of interest is the loss distribution function that assigns probabilities to default related losses over some time horizon. Assuming $\beta_i = \beta, C_i = C$ and $R_i = R$ the loss L relates to the number of defaults k by

$$L = \frac{k}{I}(1-R)$$

where I is the number of entities. The number of defaults on a portfolio is binomially distributed (individual returns R_i were independent given α) given α, so the individual conditional default probability is

$$q(\alpha) = \Phi\left(\frac{C-\beta\alpha}{\sqrt{1-\beta^2}}\right)$$

Large portfolio approximation: When the number of entities I in the portfolio is high, the CDF for the losses gets numerically hard to work with. Instead we can use the large portfolio approximation

$$\mathbb{Q}[L \leq \theta] = \Phi\left(-\frac{\Phi^{-1}(q) - \Phi^{-1}(\theta)\sqrt{1-\beta^2}}{\beta}\right)$$

CDOs: Two types:

- Traditional: CDO is based on the payoffs from an underlying pool of loans/bonds.
- Synthetic: CDO is a "pure" credit derivative, which again can be build from traditional CDOs. The basic principle in both is to split the payments associated with an underlying pool of assets into different tranches with different seniority. Buyer gets money from seller in case of defaults. Pricing CDO tranches: K_{L_j} and K_{U_j} are attachment and detachment point of tranche $j.Z_{j,t}$ is the loss percentage incurred on tranche j at time t

$$Z_{j,t} = \min\left\{Z_t, K_{U_j}\right\} - \min\left\{Z_t, K_{L_j}\right\}$$

The spread on tranche j is

$$s_j = \frac{\sum_{k=1}^K B_0^T\left(\mathbb{E}\left[Z_{j,t_k}\right] - \mathbb{E}\left[Z_{j,t_{k-1}}\right]\right)}{\sum_{k=1}^K B_0^T \eta\left(K_{U_j} - K_{L_j} - \mathbb{E}\left[Z_{j,t_k}\right]\right)},$$

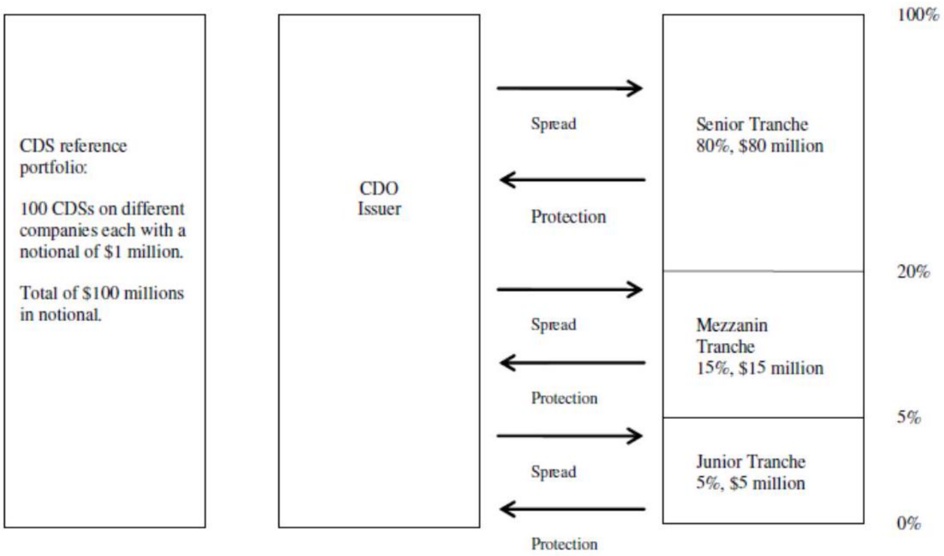


Figure: Structure in an unfunded synthetic CDO.

with η being grid size. Using LPA, pricing of a tranche boils down to evaluating

$$\mathbb{E}\left[Z_{j,t_k}\right] = \int_0^1 \min\left\{(1-R)\theta, K_{U_j}\right\} - \min\left\{(1-R)\theta, K_{L_j}\right\} \, \mathrm{d}F\left(\theta; q_{t_k}, \beta\right)$$

where

$$F\left(\theta; q_{t_k}, \beta\right) = \Phi\left(-\frac{\Phi^{-1}\left(q_{t_k}\right) - \Phi^{-1}(\theta)\sqrt{1-\beta^2}}{\beta}\right).$$

12 Fixed income issues after 2008 (CVA, pricing collateralized claims)

Idea: Let's model explicitly the credit risk and the liquidity risk in the interbank market and measure/calibrate our model.

Swap contracts are collateralized and therefore swap rates have negligible counterparty risk component. Thus, variation in swap contracts comes only from variation in the floating reference rate. The difference between a 5 year swap on LIBOR (rates for unsecured lending and borrowing between banks) and the 5 year swap on OIS (risk free interest rate proxy, rate at which banks can borrow from each other over night) comes from the difference between (risk neutral) expectations between LIBOR and OIS rates over the next 5 years. The difference between swap rates on contracts referencing different LIBOR rates are mainly due to liquidity risk (i.e. a premium is required for lending out money for longer horizons). LIBOR-OIS spread - measure of risk in the interbank market.

Pricing collateralized claims

Consider a contract with payoff/cashflow X and maturity $T. V(t,T)$ is the present value of X. Posting of cash-collateral on a continuous marking-to-market basis (100% of $V(t,T)$, best practice among financial institutions is daily). The receiver of the collateral can invest it at the risk-free rate $r(t)$ and has to pay an agreed rate $r_c(t)$ to the poster of the collateral

$$V(t) = \mathbb{E}_t^{\mathbb{Q}}\left[\mathrm{e}^{-\int_t^T r(u)\,\mathrm{d}u} X + \int_t^T \mathrm{e}^{-\int_t^u r(s)\,\mathrm{d}s}\,(r(u)-r_c(u))\,V(u)\,\mathrm{d}u\right]$$

To valuate the payoffs we discount

$$\mathrm{e}^{-\int_0^t r(u)\,\mathrm{d}u} V(t) = \mathbb{E}_t^{\mathbb{Q}}\left[\mathrm{e}^{-\int_0^T r(u)\,\mathrm{d}u} X + \int_t^T \mathrm{e}^{-\int_0^u r(s)\,\mathrm{d}s}\,(r(u)-r_c(u))\,V(u)\,\mathrm{d}u\right]$$

We denote

$$M(t) = \mathrm{e}^{-\int_0^t r(u)\,\mathrm{d}u} V(t) + \int_0^t \mathrm{e}^{-\int_0^u r(s)\,\mathrm{d}s}\,(r(u)-r_c(u))\,V(u)\,\mathrm{d}u$$

which is a $\mathbb{Q}$-martingale (all discounted price processes are MG under $\mathbb{Q}$), so

$$\mathrm{d}\left(\mathrm{e}^{-\int_0^t r(u)\,\mathrm{d}u} V(t)\right) = -(r(t)-r_c(t))\left(\mathrm{e}^{-\int_0^t r(u)\,\mathrm{d}u} V(t)\right)\mathrm{d}t + \mathrm{d}M(t)$$

With integrations by parts we obtain

$$\begin{aligned}
\mathrm{d}\left(\mathrm{e}^{-\int_0^t r_c(u)\,\mathrm{d}u} V(t)\right) &= \mathrm{d}\left(\mathrm{e}^{\int_0^t r(u)-r_c(u)\,\mathrm{d}u}\mathrm{e}^{-\int_0^t r(u)\,\mathrm{d}u} V(t)\right)\\
&= (r(t)-r_c(t))\,\mathrm{e}^{\int_0^t r(u)-r_c(u)\,\mathrm{d}u}\left(\mathrm{e}^{-\int_0^t r(u)\,\mathrm{d}u} V(t)\right)\mathrm{d}t\\
&\quad + \mathrm{e}^{\int_0^t r(u)-r_c(u)\,\mathrm{d}u}\left(-(r(t)-r_c(t))\,\mathrm{e}^{-\int_0^t r(u)\,\mathrm{d}u} V(t)\,\mathrm{d}t + \mathrm{d}M(t)\right)\\
&= \mathrm{e}^{\int_0^t r(u)-r_c(u)\,\mathrm{d}u}\,\mathrm{d}M(t)
\end{aligned}$$

Since $\mathrm{e}^{-\int_0^t r_c(u)\,\mathrm{d}u} V(t)$ is a $\mathbb{Q}$-martingale and $V(T) = X$ we have

$$\mathrm{e}^{-\int_0^t r_c(u)\,\mathrm{d}u} V(t) = \mathbb{E}_t^{\mathbb{Q}}\left[\mathrm{e}^{-\int_0^T r_c(u)\,\mathrm{d}u} X\right]$$

and conclude that the payoffs can be computed by

$$V(t) = \mathbb{E}_t^{\mathbb{Q}}\left[\mathrm{e}^{-\int_t^T r_c(u)\,\mathrm{d}u} X\right]$$

Credit valuation adjustment

Considering bilateral counterparty risk is essential in the post-crisis landscape as risky counterparties otherwise would not be able to agree upon a price. Incorporating bilateral counterparty features in a model can explain seemingly paradoxical statements. If the counterparty in a contract defaults, the market value of the contract can be lost for the outher counterpaty. In case the contract is a CDS it only makes sense to use a model that allows for a stochastic future market value (e.g. by stochastic default intensity).

CVA - "How much do I discount the price of this deal due to the fact that my counterparty can default?" (the default risk of my counterparty). Assume your own default probability is zero. Then the CVA is defined as the difference between the value of the contract assuming the counterparty is risk-free and the actual value of the contract

$$\text{Value of claim} \ = \ \text{Default-free value of claim} \ - \ \text{CVA}$$
$$\pi = \pi^{\text{Default-free}} \ + \ \text{CVA}$$

The CVA is most difficult to value when the contract has payments in both directions between the two counterparties (e.g. interest rate swap or credit default swap). An additional layer of complexity is added as we are hurt by a counterparty default when the contract has positive value for us, but not when is has negative value.

Defining the exposure at time t as

$$X_t^+ = \max\{\varepsilon_t, 0\}$$

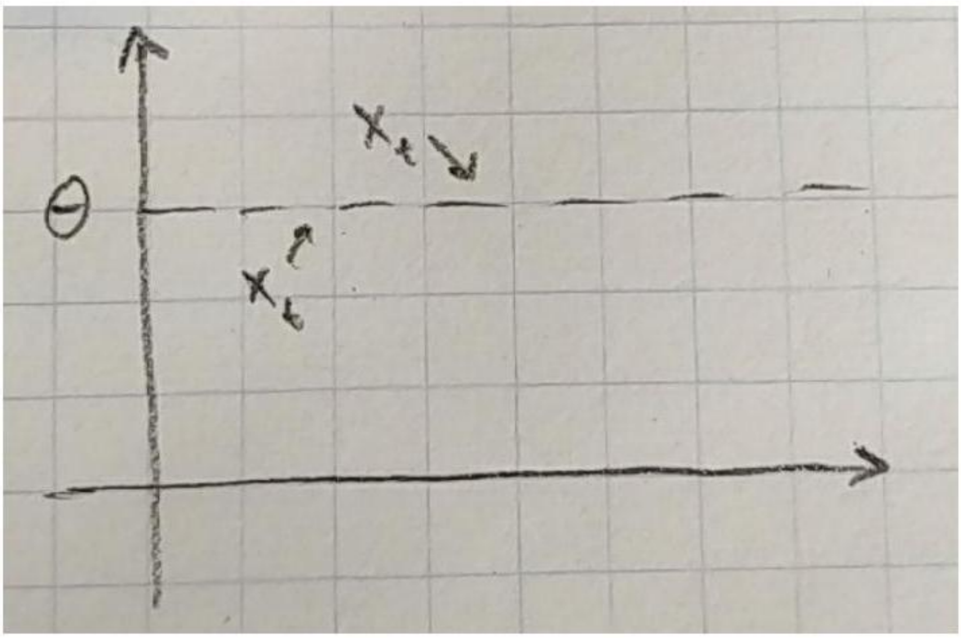
where ε_t is the close-out amount (the amount of the losses or costs the surviving party would incur in replacing or in providing for an economic equivalent of the remaining part of the deal at the time when the counterparty defaults). The pricing formula for CVA is then

$$\text{CVA}_t = -\mathbb{E}_t^{\mathbb{Q}}\left[e^{-\int_t^\tau r_s \, ds} \left(1 - R_{EC_2}\right) X_\tau^+ \mathbb{1}_{\{\tau=\tau_2 \le T\}} \right]$$

12.1 Additional stuff

Mean reversion

Ornstein-Uhlenbeck process: $dx_t = \kappa(\theta - x_t)\ dt + \beta\ dz_t$. The process is mean reverting towards θ (the mean). κ decides "how fast" it returns. β is the degree of volatility. If x_t is above θ, κ is multiplied by a negative number and the variable goes down, and vice versa.



Credit risk

Definition: The risk that an obligor does not honor his payments due to a default event.
Credit risky products are of interest to banks (and other financial institutions) because credit derivatives gives them the possibility to trade credit risk and export it out of the balance sheet. With credit derivatives it is possible to hedge the credit risk of their own portfolios.

For investors it is interesting to trade in corporate bonds and credit derivatives since the decreasing interest rate level has made corporate bonds and the higher yield on these more interesting compared to government bonds. With credit derivatives it is possible to tailor-make the credit portfolio to match the investors risk profile.

- Credit Default Swap: Suppose a bank (protection buyer) enters into a contract with an investment firm (protection seller), where it will make periodic payments to the firm in exchange for a lump sump if XYC corp. (refenrence entity) defaults during the contract term. The lump sum is denoted $(1-R)$ (assuming notional 1) and the CDS spread is determined so the contract has value zero at initiation.
- Collaterized Debt: Multi-name instrument. A collection of portfolio default swaps.
- Portfolio default swap: The protection seller covers losses in the portfolio between the attachment point and the detachment point in exchange for a periodic payment. The contract is terminated at the maturity of the contract or until the total loss in the portfolio exceeds the detachment point.
- First to default basket: A basket of reference names. The protection seller agrees to cover the loss of the first default in the basket in return for a periodic payment.
  Valuation principles: The valuation at time zero corresponds to: protection leg = premium leg.